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Modelling and simulation of a continuous beer production plant using Simulink

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SUMMARY

Beer is traditionally batch produced. However, there has been always an interest to make this process continuous, due to its economic and also environmental benefits.

In this project, a continuous brewing plant model has been developed from already published batch models. In particular, this model considers and links mashing and fermentation steps. For it, already existing forefront brewing technologies have been used as inspiration. The conversion has been performed by placing tanks in cascade for mashing and for fermentation, and by using the chemostat concept. The results allows not only to study the system behaviour but also to provide design parameters.

Besides, an overview of the brewing process and of continuous brewing developments, advantages and disadvantages is exposed.

Keywords: Beer, Brewing, Continuous brewing, Fermentation, Mashing, Continuous fermentation, Continuous mashing, Dynamic model, Kinetic model, Bioprocess modelling, Modelling, Simulation, Simulation-Based Design, Simulink

DEDICATION

To my family. To my friends. To Anni.

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1. INTRODUCTION

Beer is the most consumed alcoholic beverage in the world [1] and, consequently, it has a huge socioeconomic impact. European countries are in the top of consumption per capita rankings [1], and the European Union is the second largest producer, being China the first one [2].

Among the economic impact of beer there is its contribution to employment of 2.6 million people and of €60 billion to GDP in European Union, although both of them decreased by a 33% and a 25% respectively in 2020 due to Covid-19 impact [3]. In Spain, the job contribution was of 399,000 people in 2019, and represented the 1.5% of Spanish GDP in the same period [4].

Moreover, beyond the economic impact, there are other concerns in the European beer industry, such as the protection of water resources, the reduction of CO₂ emissions, the promotion of a responsible consumption, and health and safety issues [2]. This of course includes the Spanish sector, which assumed forefront environmental commitments in 2019 [4].

1.1. Motivation

Despite continuous brewing has aroused interest for many years, in recent years the interest has increased, being the last development a free cell fermentation system by Müller-Auffermann in 2013. Its interest resides in its economical benefits, but also in its less environmental impact compared with traditional batch processes. In fact, the European Commission embrace changing from batch to continuous fermentation systems as a green action to consider [5].

Among the years, several batch models for different brewing stages have been published. However, there has not been found any model adapted to a continuous beer fermentation free cell system, which is the type of system of the Müller-Auffermann's system. Also, any model for continuous mashing have been found, and neither a model linking both of the stages.

Consequently, due to the lack of these models, it is found an interesting research to study and develop them. Among others, its impacts could be to study the feasibility and design parameters, to develop control systems, to monitor the correct operation of the plant...

1.2. Goals

The aim of this project is to develop a continuous brewing system model, linking a continuous mashing system with a continuous fermentation one.

The lack of possibility to perform an experimental set up constraint the model to be just a theoretical one. Therefore, there is intrinsically a goal in this constrain, which is to build it from already published batch models that can be adapted to a continuous operation.

The purpose of modelling the system are to perform a later simulation to study its behaviour (its feasibility and its response to input variations); and to perform a Simulation-Based Design.

However, besides the goals of this project, the model can be used for other purposes: it can be a base for future more complex models, it can be used for develop control and automation techniques., to track a good process performance...

1.3. Methodology

The methodology that has been followed in this project is theoretically based. Therefore, a thorough review on the state of the art has been done prior to develop the final model.

After understanding the brewing process characteristics, suitable batch mashing and fermentation models that have been already published have been converted into continuous ones.

Finally, iterative simulations have been done with Simulink to find an steady operational point so some design parameters can be proposed. The dynamics of the systems have been also studied by exciting them with different signals.

1.4. Summary

The project is organized in such a way to follow an organized contents order. That way, ideas and concepts are exposed in a way to make understandable further sections.

First, in Chapter 2 an overview of the brewing process is given, so the brewing concepts are settled. It contains all the main operations carried in a brewery and its purpose.

Then, in Chapter 3, the concept of continuous beer production is presented. Differences between batch a continuous brewing are exposed, as well as advantages and disadvantages. Also, a review of the different continuous brewing stages evolution is carried out, including the technological differences between batch and continuous methods.

After having a general idea of a brewery operations, and, specifically, what are the continuous technologies and why to use them, the modelling step begins in Chapter 4. In

this chapter, a glimpse of biotechnology systems modelling is given to better understand the scope of the further actions. After a brief introduction in which biotechnology engineering is presented and the reasons of what a biotechnology system model can offer, the models of the project are explained. As the continuous models are adapted from already published mashing and fermentation models, the original batch models are explained as well as the conversion into continuous ones. For it, it is detailed the main concept that allows this conversion: the chemostat.

Later, once the models are built, the state-space concept is exposed in Chapter 5 as an widely used form of representing dynamic models. Consequently, the previously modelled systems are written in state-space form and linearized when needed.

In Chapter 6, the simulation of the system is performed after explaining some simulation insights, as well as the simulation purpose of this project. The simulation outputs are commented and some insights of the system are pointed. Moreover, a glimpse of why Simulink is chosen as the project simulation environment is justified, and the Simulink layouts are shown. Also, the model validation step is explained as an important step to do after modelling and simulating a system, although it has not been possible to perform in this project. Limitations of models are also detailed.

The results are studied in Chapter 7, in which the design parameters obtained are analysed, as well as some general aspects of the model.

Finally, in Chapter 8 the whole project conclusions are summarized, pointing the achievements of the project, and suggesting what future researches can be done to continue this project development.

The MATLAB code for setting the Simulink parameters values is attached in Appendix A, at the end of the document.

2. BREWING PROCESS OVERVIEW

Beer is a beverage made of water, malted cereals (commonly barley malt, but also others as wheat malt), yeast, and hops; although other non-malting adjuncts can be added to reduce production costs of beer or change its properties. However, in this project it is assumed the use of only these four main ingredients.

The first step in the beer production process is malting, despite this process is not performed at the brewery. The main purposes of this step are "to degrade the grain's proteins, create the enzymes and modify the starches needed for the brewing process" [6], being barley malt the main worldwide starch source for brewing [7]. It is relevant to mention that malts are a mixture of grains, and, consequently, the heterogeneity of differing properties can lead to problems in brewing [8]. Overall, the process consists in germinating the cereal to favour the accumulation of enzymes, sugars and other soluble materials [9]. First, impurities are removed from cereal grains; then, grains are hydrated by steeping (soak) them in water [8]. After, the grain germinates to be later dried or roasted in a kiln, with the purpose of terminate the growth of the plant, reduce the moisture for grain storage, conserve enzymes developed during malting and develop taste and aroma properties [7].

Secondly, the malt arrives to the brewery, where the actual beer is going to be brewed. The scope of this project is limited to the brewery operations starting since this point.

The first task performed in the brewery is the grinding. The grist is created by milling the malt grain to extract the inside of the malt and ease next extraction and dilution steps [10]. The grist size composition has an impact into the lautering properties, so the mill setting is adjusted to the mashing and wort separation systems [7]–[9].

Once the grist is obtained, it has to be mashed. In the mashing process, grist is mixed with controlled warm and hot water "to form a porridge-like mixture", this is, the mash [6]. The malt components are dissolved by enzymatic, physical and chemical processes [10]. Depending of the style to be brewed, local tradition, quality of the malt and equipment different mashing processes can be used [6].

After mashing is completed, it is time to perform a solid-liquid separation in a step called lautering. Thus, once the separation is completed, wort is the name that the obtained liquid receives. The wort is a sweet liquid in which starches have been transformed into sugars [6] in the mashing process, and obtained after lautering. The most common systems are the lauter tun and the mash filter [10]. Lauter tuns have a perforated false bottom, and rotating raking machines for cut through the mash and remove spent grain [10], [11]. In mash filters, mash is forced to pass through a vertical filter made of synthetic material frames to separate the liquid (wort) from the solid (spent grain) [10], [11]. After a first wort is obtained, grain is rinsed with fresh hot liquor (brewing water) in a

process call sparging to enhance the sugar extraction efficiency so sugars are extracted from residual solids [6]. With the use of two-vessel systems (mash tun and lauter tun) a higher speed of processing is achieved because while one vessel is mashing the other can be lautering [11].

The next step is to boil the wort to achieve several objectives [12] that are exposed subsequently. Two sub-stages can be identified in the boiling: hot holding and evaporation [10]. Several chemical reactions take place during hot holding, as development of aroma, color and dissolution processes, sterilization and inactivation of enzymes, as malt enzymes, to prevent, for example, uncontrolled over-fermented beers [10]. During evaporation, undesired aroma substances are removed [10].

During boiling hops are added. At the beginning they gave beer preservatives properties, but giving bitterness and flavour are the main reason of the current use [6], [13].

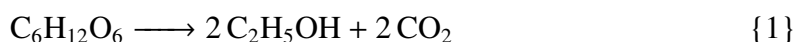
Subsequent to boiling, wort has to be clarified. The step is described according to [12]. The wort should be bright (absolutely clear) but hops and flocs of trub remains in it. The most employed method is centrifuging the boiled wort in a whirlpool tank, where the solids are settled in the centre of the base of the vessel.

Then, the wort has to be cooled until the pitching temperature, that is, the temperature in which the wort is going to be inoculated with yeast, usually about 15-22°C for ales and 6-12°C for lagers. To stop chemical reactions and minimize chances of microbes growth, it has to be performed rapidly. The heat obtained from the wort is recovered through heat exchangers, for example, to be used in previous steps as mashing.

Also, the wort "must be charged with oxygen to an appropriate level" [12] because in the first stage of fermentation the yeast needs a certain critical amount of dissolved oxygen [6], [12], [14]. Aeration can be performed with sterile air or directly with oxygen [12], [14].

The pitching is the first step to enable the fermentation [6], [15]. As the yeast cells are rapidly in contact with the wort nutrients, pitching is performed continuously into the cold wort flow [14]. Fermentation starts.

Fermentation is probably the most critical step in the brewing process. This is because beer is a by-product of yeast growth, so the brewer task is to control the metabolism and growth of yeast [14], [16]. The control is complex as several biochemical reactions occur, and wort is not completely characterized [17]. The main reaction that occurs during fermentation is the conversion of sugars into ethanol and carbon dioxide by yeast cells, although other reactions carried by yeast leads to the production of compounds which influence in aroma and flavour [6], [14], [15], [18]. According to [19], it can be expressed chemically, in a simplified way, as:



This is an exothermic process, therefore the fermenter shall be equipped with a cooling system to extract the heat generated and enable the temperature control [18] as the fermentation temperatures are between 16°C to 22°C for ales and between 9°C to 14°C for lagers [6]. Sediments generated during fermentation are going to settle at the base of the vessel or at the top of the fermenting wort, depending of the yeast type used [14], [17], [18]. Most world brewed beer are lagers, that are brewed with bottom-fermentation yeasts [18].

After the completion of primary fermentation beer has to be matured [20]. Traditionally a second fermentation occurs during maturation due to the small amount of yeast remaining [20].

Maturation can also include other beer treatments to adjust colour, flavor, and aroma, with the addition, for example, of caramel [20].

Simultaneously to maturation, clarification occurs, which includes "the removal of yeast and the sedimented protein" caused from the beer stabilization techniques [20]. It is the tank bottom which allows clarification, where yeast and other materials settle [14]. Tank bottoms can represent 1-3% of the total volume of beer [20] so, as beer of the tank bottom does not have the same properties, it is often added again to the pitching stage [20]. Although the costs of this operations can be high, it is often more performed in the developed world, where treating effluent costs are high [20]. Clarification can be also performed with centrifuges [20].

Moreover, to prevent gas release, beer should be kept cold and under the right pressure of carbon dioxide conditions, although carbonation can be applied in this step to reach the desired carbon dioxide level [20].

Finally, filtration is the last step before packaging, being beers properly matured and stabilized easier to filter [20]. The step is performed to give beer a brilliant clarity and preserve beer appearance [21]. This is reach by passing the liquid through a porous medium, so solids are removed [6]. A consequence of filtration is also stabilization, which detach potential turbidity formers, mainly proteins and tannins [15], [21].

As a final relevant step, packaging should consider three main features: to prevent air getting into the beer, to control filling pressure and temperature (<2°C) to keep CO₂ dilution, and to maintain a thorough cleaning [22].

To conclude, although it is not an actual step of the brewing process, it is important to highlight the importance of the cleaning operations. At the end, it must be considered as an integral part of the process if a trouble-free operation is desired [23], [24]. Cleaning has three purposes: remove soiling that can pollute products or affect plant operations and disinfect to prevent microbial spoilage [23]. Automatic cleaning in place (CIP) is widely used to clean several interior surfaces of the brewery equipment, including tanks, heat exchangers, pipes...[6], [23], [24]. Thus, cleaning operations must be considered, not only in the design of the plant, but also of the equipment shape and material, as well as

special care to ensure that cleaning fluids and beer streams do not cross [23], [24].

3. CONTINUOUS BEER PRODUCTION

3.1. Introduction to continuous brewing. Advantages and disadvantages.

Despite beer has been traditionally batch produced, the development of science and technology has led to the possibility of producing it continuously, with its consequent benefits and disadvantages.

The main difference between batch and continuous processes is that, while batch systems operate intermittently, continuous systems are designed to operate without shut downs [25].

Generally in process industry, continuous operations can lead to economical benefits in large scale productions [25], as they eliminate load and unload stages, reduce manpower labour, can be automated and a higher product evenness is obtained. However, their start up is complicated, they are not flexible, and need a qualified manpower labour and uniform raw material. In addition, not only economical benefits are achieved, but also environmental, as a higher resources efficiency is achieved with continuous processes [26].

In the beer industry, continuous brewing has had the highest interest during the period 1955 to 1975 although it did not have the expected results [11]. It was in the late 1950's when Morton Coutts of Dominion Breweries (currently DB Breweries) innovated with the continuous fermentation concept [27]. Fermentation is the most critical step in achieving a continuous brewing process, so the improvements in this step are the key to continuous brewing development. It is theoretically possible to make a continuous fermentation process in which wort is rapidly converted into beer as the yeast is continuously active, that is, the lag phase of a batch fermentation is not occurring [28]. Thus, the interest in continuous brewing has increased in recent years with the develop of continues fermentation with immobilized yeast [11] and, more recently, with Müller-Auffermann's free cell system model [29]. The reason of developments in the whole continuous brewing process is because it is more advantageous if all steps are continuous [11], [30]. Thus, not only successful advances has been reached in continuous fermentation, but also in other stages such as continuous mashing and boiling with Meurabrew by Meura [26] or continuous lautering with Nessie system from Ziemann [31]. Continuous brewing stages details and history will be described further.

In addition to the general characteristics of continuous processes, there are other specific continuous brewing features. These are hereafter explained, and all of them summarized in Table 3.2.

Concerning energy and utilities brewhouse consume, a continuous production reduce peak-consumption and energy losses [26], [27], [32]–[34]. During batch production, all

utilities have a specific high consumption period which consequently generates peaks, specially in steam and cooling liquid [26]. Concerning energy losses, as the vessels are not continuously operating, vessels are cooling down between different batches [26].

Waste is also reduced by using a continuous brewing process instead of a batch one [26], [33], [34]. Rinsed water used between batches to clean the vessels not only increase water consumption and peaks [32], but also waste water [26].

Other utilities and waste are reduced as less cleaning is needed due to a longer time stable equilibrium is reached in continuous systems [32]–[34]. According to Müller-Auffermann calculations, 94% of cost savings in cleaning can be achieved [33].

However, achieving a long time stable equilibrium increases the risk of microbial contamination [27], [32], which is one of the disadvantages of continuous processes. Nevertheless, an uniform beer quality is expected as long as the process keeps stable [11], [26], [32], [33]. Also, mutations of organism risk is increased due to aging and long-term stress [33]. Therefore, if a deviation occurs, qualified personal must be on-site 24 hours a day [32], [33]. In spite of this, an increase in labor productivity and a reduction of personal costs are obtained with continuous processes [33] because only one fermentation must be controlled [27]. In addition, for process stability, it is needed a constant raw materials quality [33].

A continuous production reduces the number of vessels used and, consequently, the brewhouse space requirements [26], [27], [32], [33]. Moreover, as utilities consumption peaks are reduced and less flow is managed, smaller pipes diameters, valves and pumps are required [26]. Logically, this leads to a cost of capital reduction [27], [33].

A comparison between a batch and a continuous brewery utilities, with the same total brewhouse production requirements, provided by Meura [26], is shown in Table 3.1.

Batch brewhouse	Continuous brewhouse
12 brews/day of 400 hl cold wort	200 hl/h cold wort
Mash pump: 1500 hl/h - 15 kW	Mash pump: 180 hl/h - 5,5 kW
Wort pump: 3600 hl/h - 30 kW	Wort pump: 225 hl/h - 4 kW
Steam peak flow: 14 T/h	Steam peak flow: 3 T/h
Water peak flow: 650 hl/h	Water peak flow: 220 hl/h
Electricity installed: 375 kW	Electricity installed: 250 kW
Electricity peak: 300 kW	Electricity peak: 200 kW
Cooling power peak: 4650 kW	Cooling power peak: 2200 kW

Table 3.1. COMPARISON BETWEEN A 3 MILLION HL FINAL CAPACITY BATCH AND CONTINUOUS BREWHOUSE CONSUMPTION [26].

In addition, beer losses are reduced [32], [34], the amount of pitching yeast needed is less [32], and CO₂ recovery is improved so less greenhouse gases are released [5], [27],

[33].

However, a steady demand is preferable to choose a continuous fermentation system [11], being an inconvenience the poor flexibility with respect to sales fluctuations [33]. Nevertheless, flow rates variations according to demand are done, at least, in DB Breweries [27].

To sum up, not only economical benefits but also environmental could be obtained by implementing continuous fermentation technology. These environmental benefits are aligned with current and future beer industry trends of energy efficiency and sustainability [33]. This can be also reflected in the Green Best Practice Community from the European Commission, who includes moving from batch to continuous fermentation systems as a green practice to consider [5].

ADVANTAGES	DISADVANTAGES
No load and unload stages Manpower labour reduction Can be automated Labour productivity increase Peak-consumptions are reduced Energy savings Less water consumption and waste Less cleaning products consumption Only one process is controlled Less number of vessels Less space requirements Smaller utilities needed Capital reduction Beer losses are reduced Less pitching yeast needed Improved CO ₂ recovery	Complicated start up Lack of flexibility Qualified manpower needed 24 hours a day Uniform raw material required Higher microbial contamination risk Higher organism mutation risk

Table 3.2. ADVANTAGES AND DISADVANTAGES OF CONTINUOUS BREWING PROCESSES. OWN COMPILED.

3.2. Continuous brewing stages developments

Improvements in the whole brewing process have been made to achieve a continuous brewing during the last 100 years [32], [34].

Mark Strobl makes a review of the advances reached in each step of continuous brewing [32]. It is hereafter explained and complemented, while in Table 3.4 a comparison between batch and continuous processes steps is compiled.

The first step, milling, is actually a continuous process where the malt silo and the milling body act as buffers before and after the milling [32]. Capacities of these buffers can be reduced if the next step, mashing, is continuous too [32].

Nevertheless, it is easy to perform continuously the mashing step with heat exchangers varying temperature and speed of the mash [32]. Meura, which has achieved a continuous brewing plant at an industrial scale in 2007 (Meurabrew), utilizes different mash vessels to keep mash at constant temperature during a specific time, while a continuous flow passes through the vessels and their respective filters [26], [32], [35], [36]. The sketch of the process can be observed in Figure 3.1.

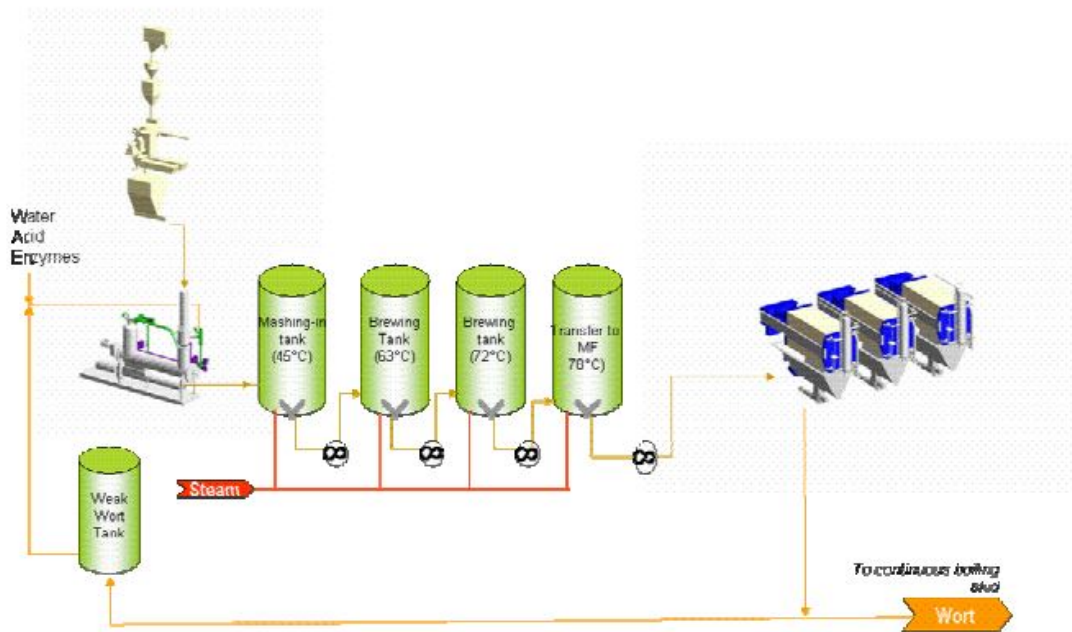


Fig. 3.1. Meurabrew continuous mashing and filtering system [26].

Next, continuous lautering is achieved with vacuum rotatory filters [37] or decanters, to withdraw the indissoluble particles from the wort [32]. Several processes of continuous wort production have been launched. Harsanyi developed in 1968 a system to continuously centrifuge the mash, rinse the dehydrated solids with water to remove them automatically and, finally, force the wort to further clarifications before brew it in the kettle [32]. More recently, in 2016 was introduced the Nessie system, from Ziemann [32], [38]. In this system that combines a continuous lautering and mash filtration, the separation of wort and spent grains is performed by four rotatory disc filters in cascade [31], [32], [38]–[40] and a 34% time saving is achieved [31], [32], [39].

Concerning the following stage, continuous boiling methods have been developed since the second half of the 20th century with high boiling temperatures [32]. However, more recently, the Meurabrew system has also successfully reached a continuous wort boiling operation [26], [32], [35]. An outline of the process can be seen in Figure 3.3. First, the wort is heated until the boiling temperature, and hops are homogenized [26]. If natural hops are used, a hop strainer must be included [32]. To ensure an even mixing an



Fig. 3.2. NESSIE continuous lautering system [38].

external agitator is used [26], [32]. Then, the wort is continuously clarified using a wort settling tank as an alternative to a whirlpool, which is not possible to use in a continuous operation [26]. In this stage, non-oxidized hot trub is recovered to use it again in the mashing [26]. It is important to perform this task before stripping to avoid trub particles soiling the column [32]. Then, in the stripping column, volatile particles are removed by a counterflow clean steam [26], [32]. Afterwards, wort is cooled down through two duplicate wort coolers [26], [32]. Thus, one can be cleaned while the other is operating [26], [32].

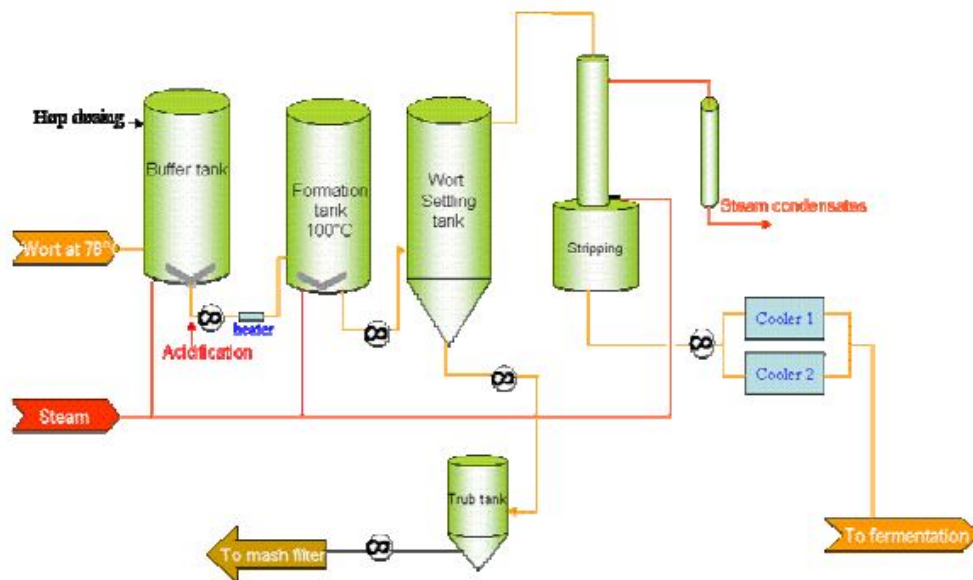


Fig. 3.3. Meurabrew continuous boiling and clarification [26].

Afterwards, fermentation, as the most critical stage to achieve a continuous brewing process, has experienced different periods of more and less interests and advances. Continuous fermentation was already tried before 1900, and at least five different systems were

detailed before 1906 [18]. Among these systems, the idea of implementing a cascade of partly fermented beer from one tank to another, which would be used later appeared [18]. The interest was renewed in the 1950s and 1960s, due to an increase in beer consumption, and a hope of advance in this area raised again due to the enhanced industry knowledge and electronics and control advances [18].

Despite different companies attempted, finally only Morton Coutts of Dominion Breweries (currently DB Breweries) succeeded establishing their system in New Zealand in the late 1950's [18], [27]. The New Zealand system includes a hold-up vessel, two stirred fermenters and a yeast separator [18], [27]. The purpose of the hold-up vessel is to aerate and mix the wort with yeast and recycled beer obtained from the first fermenter [18], [27]. The system is deeper described hereafter according to [27]. Because of the high nutrients levels, the pitch of new yeast into the wort can be stressful for the yeast, so by introducing partially fermented beer, nutrients levels decrease and fermentation begins earlier. Moreover, the introduction of partially fermented beer also leads to a pH reduction and an ethanol increase, creating an unfavourable environment for any competing micro-organisms. Also, the recycled yeast is still in an active fermentation stage, which avoid a lag before fermentation starts with the new introduced yeast. Then, the first continuous fermentation vessel is the main of both two, with around a 60% of the continuous fermentation volume; while the second one is more focus on the fine-tuning of the finished fermented beer, assuring the ethanol desired amount. After, in the yeast separator, surplus of yeast is washed and sold. Finally, the beer is pumped to a maturation vessel.

However, the New Zealand system is only a continuous fermentation system, but it does not use continuous mashing and wort boiling. Wort is batch produced and then storage in cold storage tanks at 0°C (the wort does not freeze due to its high sugar content). Then, wort is fed continuously to the system from the storage tanks. Up to one year continuous runs have been reached [18].

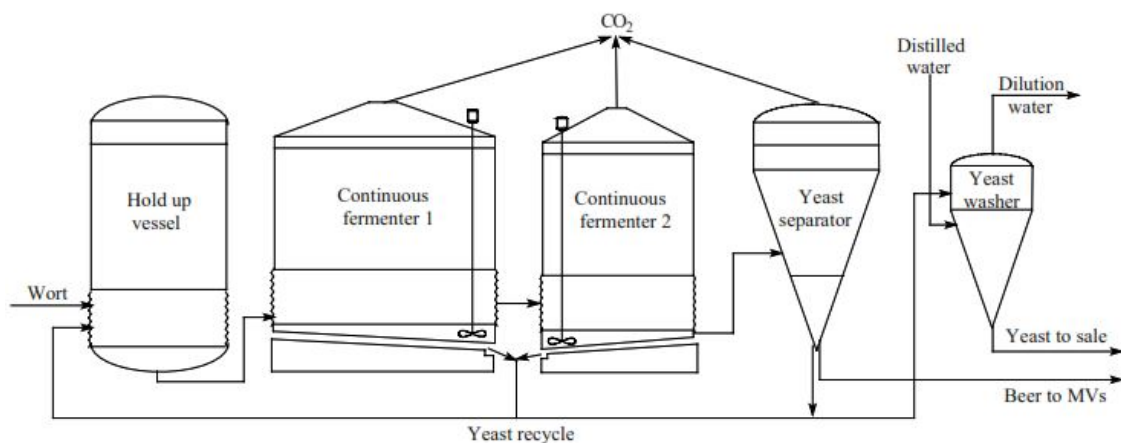


Fig. 3.4. New Zealand system [27].

A similar approach was operating in the United Kingdom in the 1970s, achieving an output of 32,000 hl per week, although it did not reach a commercial feasibility [18]. It

comprised also two stirred fermenters in series and, for collecting the yeast, a sedimentation vessel [18], as it is described in Figure 3.5. However, yeast and wort recycling was not done in this approach.

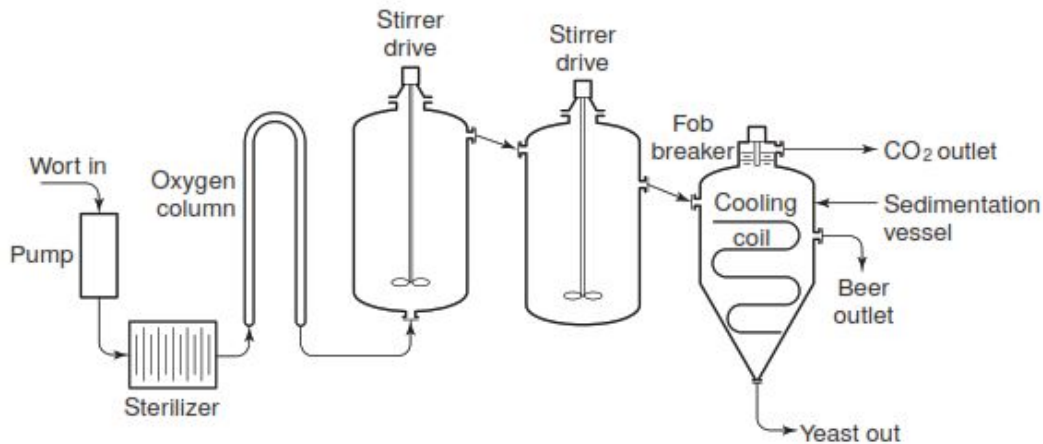


Fig. 3.5. Stirred tank continuous fermentation [18].

More recently, in the late 20th century, continuous fermentation experienced an interest renaissance with the development of immobilized yeast reactors [16], [18]. The aim of this technique is to fix the yeast in a permanent position, so a high yeast concentration is obtained, yeast are not free and therefore an heterogeneous fermentation is achieved [18]. The purpose of obtain an heterogeneous fermentation is because a high yeast concentration is needed for perform the continuous fermentation, but, on the other hand, yeast grows slow in high concentration environments [18]. The biggest step in this technology was done with the development of commercial immobilized yeast reactors in Finland [32]. Meura Delta has also make immobilized yeast reactors [41], [42]. In 2001, Virkajärvi proved the feasibility of immobilized yeast systems reaching a 14 months operating period in the experiment set up [42]. However, immobilized yeast systems did not spread successfully due to engineering problems and unrealized cost advantages, although recent developments and understanding about immobilized yeast can help to revitalize research again [43].

The latest advance has been done in free cell systems in 2013, by Müller-Auffermann, with the Technische Universität München [32] and Krones [29]. Four tanks are disposed in series, leading to a cascade reaction [32]. The system can be implemented in current set ups of cylindroconical vessels, equipping them with a central downward-facing pipe with open bottom [34]. Thus, tanks can be fill and empty continuously from the top part, while yeast and other sediments can be removed and, if required, re-used [32], [34], which lower the risk of yeast mutations [33]. A diagram about the operation of the system can be observed in Figure 3.7. One of the advantages of this system, apart than it can be implemented in current installations, is that it can be seen as a continuum of batch cultures [32], which makes it easier to model and manage, as it is operated practically identically to the traditional process [34].

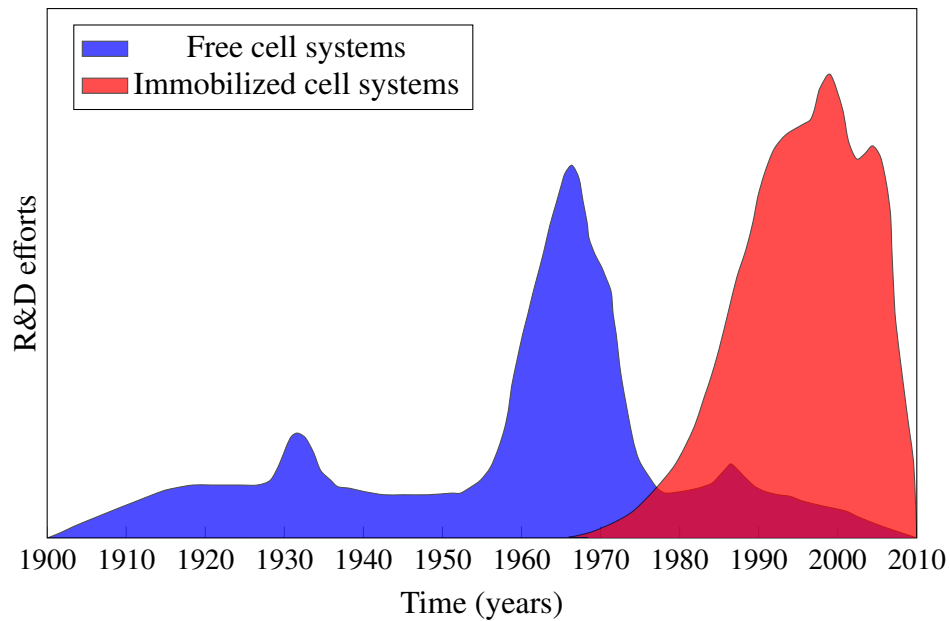


Fig. 3.6. Research and Development effort in continuous fermentation processes [33].

Müller-Auffermann argued that, although lately research has been focus on immobilized yeast systems, actually few free cell systems were tested around 60 years ago [33]. Thereby, his development has gone against the trend that can be observed in Figure 3.6. Advantages and disadvantages of free cells and immobilized cells systems are exposed in Table 3.3. Anyway, Müller-Auffermann's system has fulfill important demands such a nearly identical treatment of yeast to classical batch processes, safety against infections, simple usage or an easy integration to current plants [33]. Consequently, the system can be easily simulated and operated, as it is similar to well known batch processes [33].

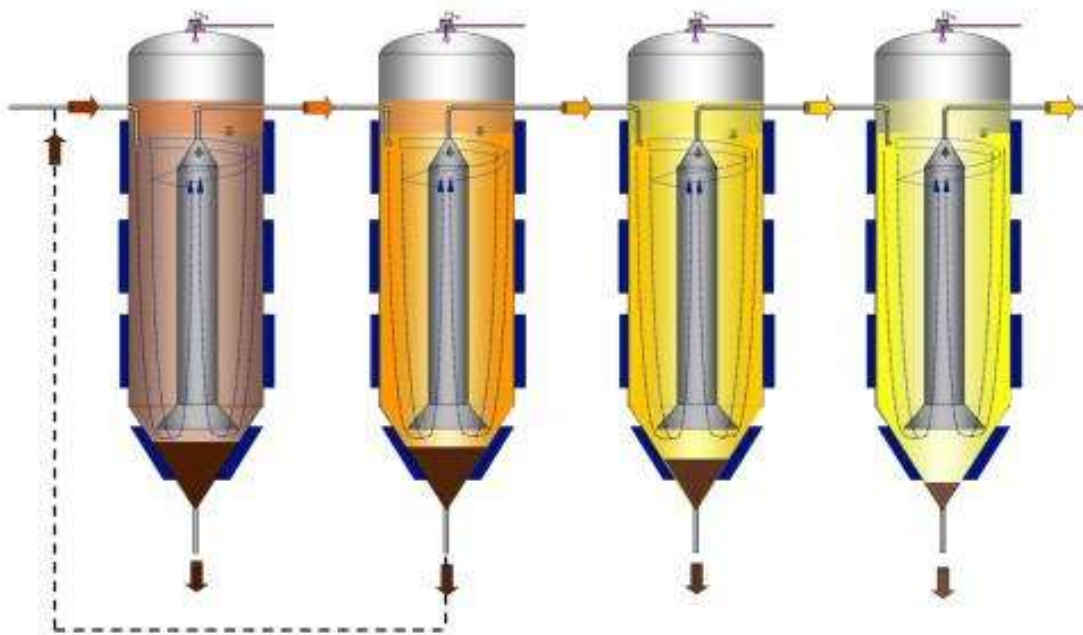


Fig. 3.7. Müller-Auffermann continuous fermentation system [34].

FREE CELL SYSTEMS		IMMOBILIZED CELL SYSTEMS	
Advantages	Disadvantages	Advantages	Disadvantages
Yeast cycles can be reproduced	Larger reactors	Small reactors	Long term stress for yeasts
Aged yeast can be discharged	More produced biomass	Less produced biomass	Yeast ages in system
Lower risk of permanent infections	Rather complicated systems	Uncomplicated systems	Higher risk of yeast mutations
	Perhaps lower efficiency	Perhaps higher efficiency	Higher risk of permanent infection

Table 3.3. COMPARISON BETWEEN FREE CELL FERMENTATION SYSTEMS AND IMMOBILIZED CELL FERMENTATION SYSTEMS[33].

Afterwards, the clarification is done with centrifuges, decanters [20], [32] or through a rotatory vacuum filter [20]. Concerning filtration, a cross-flow filter can be the solution[32].

Finally, the filling stage can be performed continuously by modern machines, which can adapt their speed to keep the continuous production [32]. In fact, "for optimum efficiency the filler must run continuously" [22].

STEPS IN BEER PRODUCTION		
Process	Batch	Continuous
Milling		All mills
Mashing	Mash tun	Tube heat exchanger
Lautering	Lauter tun, mash filter	Centrifuge, decanter, Nessie system
Wort boiling	Copper kettle	Heat exchanger
Wort clarification	Whirlpool, flotation	Decanter, centrifuge
Wort aeration	Flotation	Inline nozzle
Fermentation	Fermentation tub or tank	Immobilised yeast, Müller-Auffermann's system
Maturation	Storage tank	Centrifuge
Filtration	Dead-end cave filtration	Crossflow filter
Filling	Stationary fillers	Rotary-type machines

Table 3.4. COMPARISON BETWEEN BATCH AND CONTINUOUS STEPS IN BEER PRODUCTION. ADAPTED FROM [32].

4. PHYSICAL MODEL OF THE SYSTEM

4.1. Introduction to biotechnology systems modelling

Biotechnology is the usage of microbiology, biochemistry and genetics; and, in its application to industrial processes, control and process engineering are required [44]. Despite it is an area in continuous improvement, it has been developed for thousands of years: biotechnology origins are found at the beginning of human history, with the necessity of understand fermentation processes [44]. It is estimated that alcoholic fermentation was discovered around 10.000 B.C. by Sumerians, and Egyptians found how to bake with yeast around 5.000 B.C. [44].

In biotechnology industry, systems modelling can help to understand, predict and optimize the factories and processes [45]. Thus, strategies evaluation and improvement can be done with valid models [46]. Also, modelling can be crucial to optimize and control bioreactors and fermentation processes [46]. Other modelling purposes include minimizing undesired by-products formation, a product quality increase [46] or minimizing energy consumption [47]. Thus, modelling is essential for a later control of the system and a good supervision and operation of bioreactors [48].

An appropriate modelling procedure, which will lead to the mathematical model of the system, must be selected [49]. Consequently, in the fermentation modelling, an appropriate model has to be chosen among different modelling approaches, as it will be discussed later in Section 4.2.

The first step to model a system is to identify its fundamental properties [49]. In this project, a continuous fermentation beer production plant is modeled. Thus, even if the whole beer production process stages have been explained in previous chapters, the present model is simplified, and considers the most important steps in a brewery: mashing and fermentation. Although boiling is a relevant stage in the process too, it has not been included in this project for several reasons. First, it is not influencing the fermentation process, which is the last stage of the present model, so it can be neglected in that aspect as the main reactions that occurs during boiling are related with the isomerization of hops acids to shape the flavor, aroma and other properties, but that does not affect the fermentation. As consequence, the model is not going to be detailed in several compounds that impact these properties, but a simpler model is achieved. However, it is proposed as a future work, and the hops acids kinetic model during wort boiling in [50] is suggested as a first approach to the problem.

Once the two main stages to be modeled are defined, a research on current batch models has been done, so they have been later adapted to the continuous problem. This "new" continuous models are proposed from the modification of the batch ones, providing

to the literature models for the continuous problem that have not been found.

Hereafter, the physical models are detailed, starting from the fermentation process as it is the most critical and relevant stage to obtain a continuous beer production. Moreover, some concepts used to transform the batch model in a continuous one, as the dilution rate concept, are more suitable to be explained in the fermentation step, but are also used in the mashing one.

Later, in Chapter 5, the physical models are going to be linearized, if needed, to represent them with a state-space form to simulate them.

4.2. Fermentation modelling

Being the fermentation stage the most critical step during the brewing process, its model will be one of the key parts of the whole plant model, and a special attention is paid to it.

In biotechnology engineering, there are four process modelling methods that might be considered:

- **Physiological model.** The biological growth process physiology is expressed in a causal-consequent statements relation [48], [51].
- **Structured model.** A system of partial differential equations is used to describe the dynamic behaviour of the cells activity [48], [51], being the biomass structured or classified by intracellular characteristics which describes growth [48]. Sometimes partial differential equations are simplified into ordinary differential equations [48], [51].
- **Unstructured model.** In this model fermentation is represented by a single, homogeneously growing, organism [48], [51]. It has been the most used model for fermentation control systems [48].
- **"Black-box model".** Also known as "input-output" model, it is based in express the output variable in terms of a function of the system inputs [48]. However, it is not necessarily biologically relevant [48].

The type of modelling approach that is going to be used is the unstructured model. Thus, a fermentation as an homogeneous organism is going to be considered as the main purpose of this project is to model and control a whole brewing process. Thereby, fermentation, in spite it is the most relevant stage, is only one of all that compound the brewing process, and consequently an interest in intracellular characteristics does not exist in this project.

The fermentation technology that is going to be modelled and simulated is the one recently proposed by Müller-Auffermann, which is previously described in Section 3.2.

The reason of choosing this system is because, as it is described earlier, due to its similarities to well known batch processes it can be easily simulated [33]. Hence, previous batch fermentation models can be used. Particularly, in this project, a kinetic beer fermentation model, published by Parcunev *et al.*, and which is applicable to laboratory and industrial operations [52] is used. Moreover, it "gives reliable results with a good description of the fermentation process" [52], which is desired if the model is going to be fitted into a continuous operation. In addition, the adaptation from a batch to a continuous fermentation model has been made following the similarities found in [53].

However, the Müller-Auffermann continuous fermentation technique is not only chosen due to its facilities to be modelled due to a batch process similarity, but also, from a practical approach, due to its easy integration in current installations and its safety against infections.

The first step that is done is to define the physical system that it is going to be modelled. In this case, as the Müller-Auffermann design is taken as reference, the modelled system is going to be compound by four cascade reactors.

Once the system is defined by four cascade reactors, the Parcunev *et al.* kinetic model is adapted to it. This adaptation is described in the following lines. On it, the main players on a fermentation process are described: biomass (yeast) (X), ethanol concentration (P), and substrate concentration (S).

For a better understanding of the construction of the reference model, it is interesting to introduce the Balling Equation. Balling established a method to measure the transformation of wort sugars into ethanol, during beer fermentation, in 1843 [9], [54], [55]. Essentially, what Balling empirically based suggests is that, during fermentation, 2.0665 g of extract are converted into 1 g of alcohol, 0.9565 g CO₂, and 0.11 g of yeast [56]. Extract, which is not homogeneous and its made by different sugars and other compounds, is usually considered as an only sucrose solution [56]. The reason of why sucrose is set as the reference is because the main ingredient of the extract is maltose, which has very similar physicochemical properties, as both of them are disaccharides [56].

Considering this relationship, a mathematical model has been built by the authors, establishing the following yield coefficients:

$$Y_{X/S} = 0.11g_{biomass}/2.0665g_{extract} = 0.0532$$

$$Y_{P/S} = 1g_{alcohol}/2.0665g_{extract} = 0.4839$$

From Balling's Equation, the evolution of the substrate (extract) concentration, which is described in Equation 4.1, can be easily understood:

$$\frac{dS(t)}{dt} = -\frac{1}{Y_{X/S}} \cdot \frac{dX(t)}{dt} - \frac{1}{Y_{P/S}} \cdot \frac{dP(t)}{dt} \quad (4.1)$$

On the other hand, biomass and ethanol evolution are described in Equations 4.2 and

4.3, in which $\mu(t)$ is the specific growth rate, and $q(t)$ is the product accumulation rate.

$$\frac{dX(t)}{dt} = \mu(t) \cdot X(t) \quad (4.2)$$

$$\frac{dP(t)}{dt} = q(t) \cdot X(t) \quad (4.3)$$

As a curiosity, in 1918 Plato revised Balling's equivalences and, consequently, wort gravity is commonly expressed as degrees Plato (°P) [9]. Degrees Plato measures concentration in percentage weight/weight [18].

The way Parcunев *et al.* has determined the kinetics parameters of fermentation is with three of the most used kinetic models [52]. That is, $\mu(t)$ and $q(t)$ have been computed following three different wide-used kinetic models: Monod, Aiba, and Tiessier. According to the authors, both the Monod and Tissier patterns give a satisfactory accuracy, being difficult to choose between one of them; while the Aiba model could be simplified to the Monod one, as there is a weak product inhibition [52]. In this project, the Monod kinetic equation is chosen against the Tissier one. Monod equation is the most usual way to describe cells growth [57], indeed, the Monod equation is also chosen because continuous culturing method using a chemostat was described by him [58], [59]. The chemostat concept is going to be detailed later, in Section 4.2.1. The rates $\mu(t)$ and $q(t)$ expressed with a Monod type equation are detailed in Equations 4.4 and 4.5. On them, K_{sx} and K_{sp} are constant kinetic parameters. The values of these parameters for each one of the three kinetic models can be found in the original paper [52]. The comparison between the theoretical model and the experimental results are shown in Figure 4.1, for a free cells system fed with a 9 °P extract [52].

$$\mu(t) = \mu_{max} \cdot \frac{S(t)}{K_{sx} + S(t)} \quad (4.4)$$

$$q(t) = q_{pmax} \cdot \frac{S(t)}{K_{sp} + S(t)} \quad (4.5)$$

It is interesting to notice that Parcunев *et al.* model considers that cell growth is beginning since the pitch moment, and does not considers the lag phase. Lag phase is the time period between pitching the yeast and the beginning of fermentation, as yeast needs some time to adapt to the new environment and shift to a metabolic activity [6]. Theoretically, the kinetics models proposed by the authors should not be used in the lag phase, as the kinetics approximation is applicable for pseudosteady-state cases [57]. However, as it can be noticed in 4.1, even if the lag phase is not considered, the results are quite satisfactory. That might be because the lag phase last for about 3 to 15 hours [6], while the whole fermentation process last for about 10 days. Anyway, the lag phase produced after pitching is not an issue to consider in this project, as the continuous fermentation model that is built around the operational point, which is assumed to be kept along time.

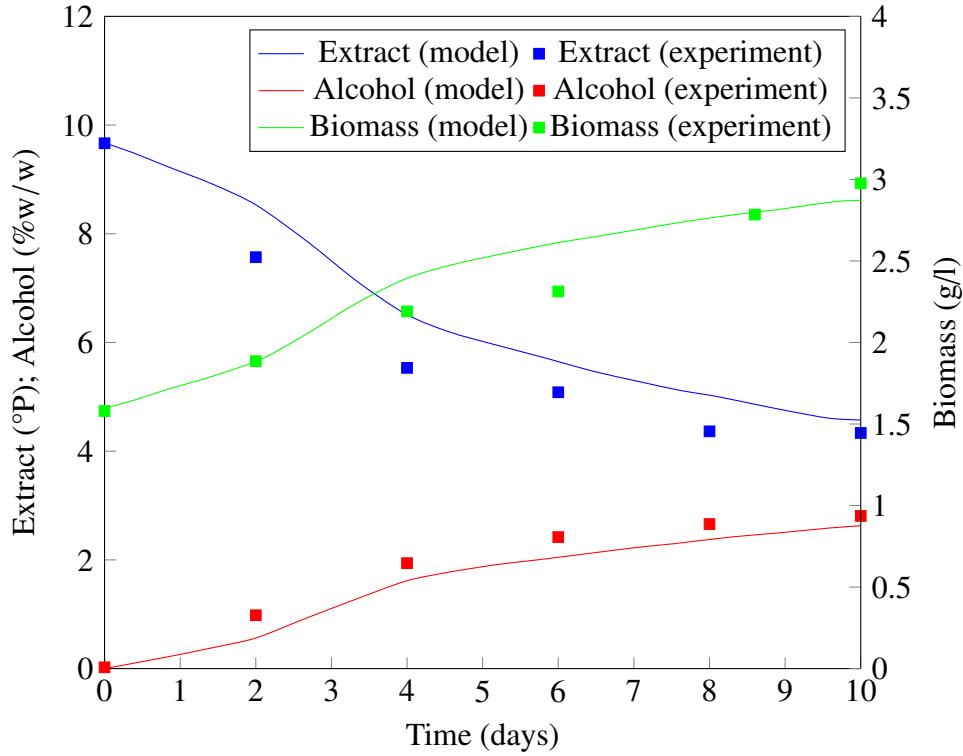


Fig. 4.1. Comparison between batch fermentation model and experimental results. Free cells system, 9 °P extract [52].

4.2.1. Fermentation model. From batch to continuous operation

Parcunев *et al.* model is built up for a batch fermentation process, and it has to be adapted to the continuous cascade system proposed in this project. Considering the continuous cascade model by Bovee and Sevely [53], a mass balance might be done for every tank. The reason of why this model is not directly used is because it is made for ethanol production and not specifically for beer production, so the parameter values might not fit to a beer fermentation process. Moreover, any parameter values are given on it. On the other hand, Parcunев *et al.* model is made for beer fermentation, and it provides actual parameters values for it.

Anyway, the mass balance that is adjusted to the present problem is made in a similar way than in the Bovee and Sevely model. For it, is important to introduce some concepts, as the chemostat one, which is the most used control technique for continuous fermentation processes [60]. A chemostat, described independently in 1950 by Monod [59] and Novick & Szilard [61], is a bioreactor in which cell growth is controlled by diluting the media by adding new media (the growth limiting factor) while simultaneously old media and cells are removed [58]. In the model previously exposed, the cell population is represented by X , and the growth limiting factor, the substrate, by S .

The dilution rate, D , relates the flow of medium into the reactor, Q , with the culture volume of it V , such that $D = \frac{Q}{V}$ [60]. In the steady state of a continuous system, which

is desired to be reached, the specific growth rate (μ) is equal to the dilution rate, that is, the rate at which the mixture is diluted [58]. This implies that the inlet flow of medium equals the outlet one [60]. Hence, it is feasible to stabilize steady-state populations of cells at different growth rates and nutrient limitation by varying the dilution rate [58]. It generally takes eight culture generations to reach the steady state [58], although reaching a perfect steady state is complicated in some systems, and, instead, it will be a pseudo-steady state [60]. The approach to the steady state in a generic single tank chemostat can be observed in Figure 4.2, where the red line represents the nutrient concentration and the blue one the cell density.

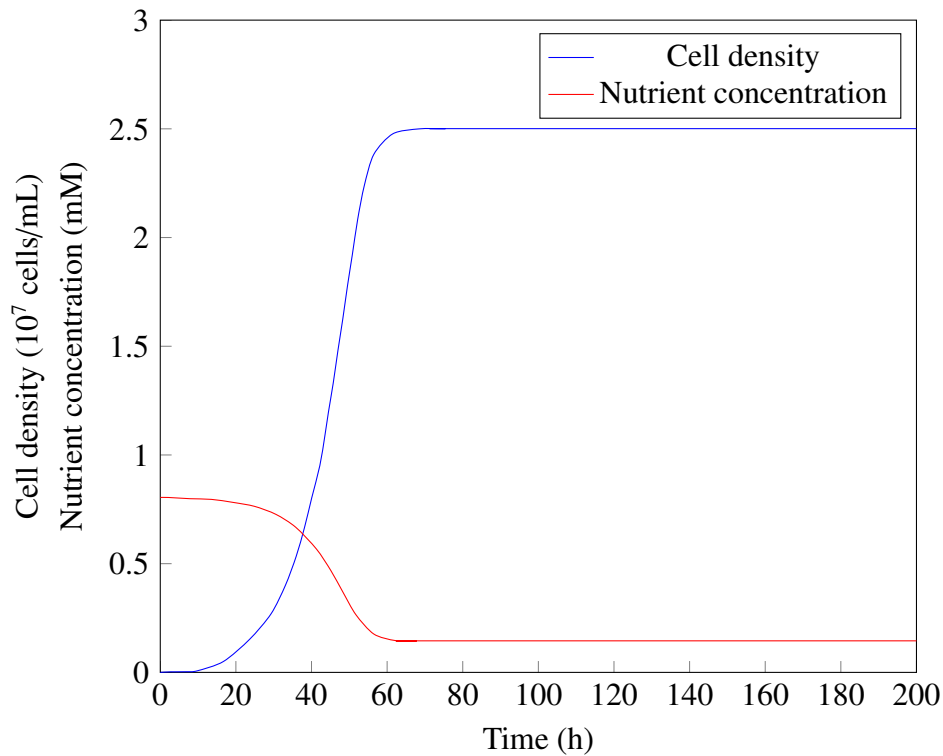


Fig. 4.2. Steady-state theoretical approach of a single tank chemostat defined by Monod type equations [58].

Consequently, once the dilution rate concept is understood, it is included in the model, considering the four tanks cascade. An outline of the whole system model illustrating the naming and linking mashing and fermentation stage is presented in Figure 4.3. It has to be pointed that it is not necessary that all the cascade tanks have the same dilution rate: all of them have the same input and output flow rate, Q , but they can have different volumes.

Therefore, the batch model is adapted to the cascade of reactors, which leads to Equations 4.6, 4.7 and 4.8, where D_{fi} represents the dilution rate of the fermentation stage in

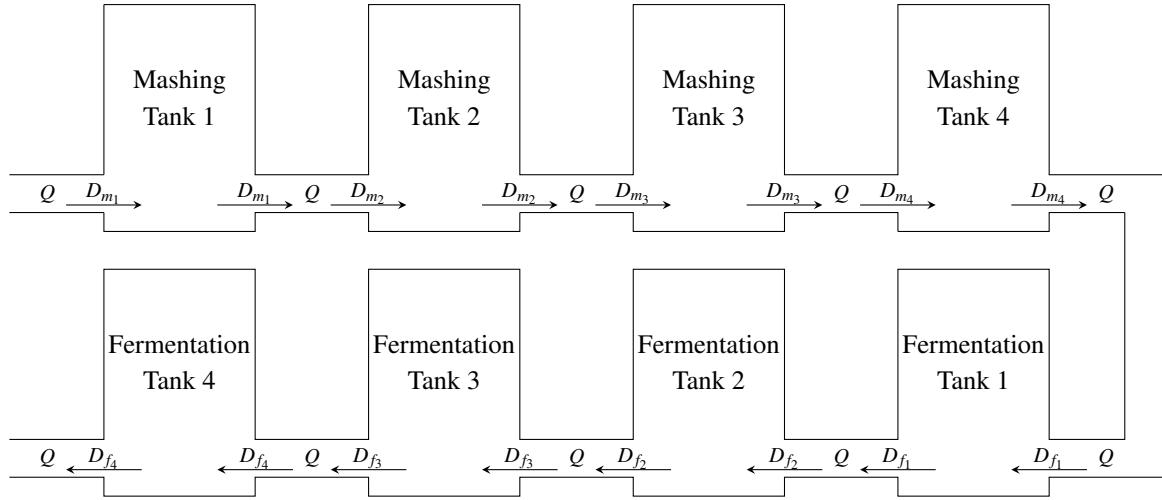


Fig. 4.3. Outline of the system model.

the tank i . In a similar way, the same approach is represented for mashing.

$$\frac{dS_i(t)}{dt} = D_{f_i} \cdot (S_{i-1} - S_i) + \left[-\frac{1}{Y_{X/S}} \cdot \left(\mu_{max} \cdot \frac{S_i(t)}{K_{sx} + S_i(t)} \cdot X_i(t) \right) - \frac{1}{Y_{P/S}} \cdot \left(q_{pmax} \cdot \frac{S_i(t)}{K_{sp} + S_i(t)} \cdot X_i(t) \right) \right] \quad (4.6)$$

$$\frac{dX_i(t)}{dt} = D_{f_i} \cdot (X_{i-1} - X_i) + \mu_{max} \cdot \frac{S_i(t)}{K_{sx} + S_i(t)} \cdot X_i(t) \quad (4.7)$$

$$\frac{dP_i(t)}{dt} = D_{f_i} \cdot (P_{i-1} - P_i) + q_{pmax} \cdot \frac{S_i(t)}{K_{sp} + S_i(t)} \cdot X_i(t) \quad (4.8)$$

It is important to highlight that, in Equation 4.1, the substrate evolution is literally expressed in terms of the derivatives of yeast and alcohol concentration; while in Equation 4.6, this can not be done. The reason is because now the derivatives of yeast and alcohol concentration depends also on the amount of yeast and alcohol entering and leaving the tank, while the original expression does not. The effect of expressing the substrate evolution in terms of the whole new yeast and alcohol evolution expressions, which includes also the incoming and outgoing flows, can be analyzed theoretically. In a steady state, the yeast and alcohol derivatives would be zero. That means that the right part of Equation 4.6 would be zero too, leading to the unreal situation in which the sugars entering the tank would be equal to the sugars leaving the tank.

Finally, it is pointed that, for the first tank, the supply of yeast X_0 and alcohol P_0 is equal to zero, as there is not any yeast and alcohol content in the wort that is being introduced into the tank. It is not necessary to pitch any yeast as there is already active yeast in the tank. The incoming sugar concentration, S_0 , is equivalent to the output sugar concentration of the mashing state C_{g4} , which is described later in Section 4.3. Moreover, the kinetic parameters are assumed the same for all the tanks, as at the end the cascade is a batch fermentation which is split in four steps.

4.3. Mashing modelling

Regarding the modelling of the mashing step, the system that is going to be taken into consideration is the Meurabrew continuous mashing system, previously explained in 3.2.

Thus, operating also with four cascade tanks, the temperature is going to be assumed constant in each tank, as it can be seen in Figure 3.1. That is, 45°C, 63°C, 72°C and 78°C consecutively. Thereby, as in the continuous fermentation adaptation has been done, a model for a batch mash production is going to be adapted.

There can be found several mashing models. In 2003, Brandam *et al.* built a model for the starch degradation during beer production mashing [62]. However, this model has been neglected because it goes too much in detail into the mashing science. It takes into account amylase enzymes, in charge to convert starch into sugars [10], [63], and models the evolution of the main different resulting sugars: glucose, maltose and maltotriose.

The kinetic model that has been considered is the one recently proposed by Tschoeke *et al.* in 2019 [64]. It is a multidimensional kinetic approach which considers the effect of several variables (temperature, pH, grain size, dilution rate and processing time) for the sugar formation, measured in terms of Brix values [64]. Despite degrees Brix represent the percentage by mass of sucrose in a solution [65], in Tschoeke *et al.* model [64] Brix is assumed as the measure for the fermentable sugars, being a good approximation of all of the different sugars present in the solution [65]. Thus, this model has been chosen because of its simplicity against Brandam *et al.* model, as it does not consider amylase kinetics and treats fermentable sugars as an unique single output. In fact, the fermentation model of the present project considers the fermentable sugars as an unique substance (the substrate), which makes worthless to consider all of them separately in the mashing stage. Additionally, the authors state that it is one of the most complete on literature [64]. Some theoretical and experimental results are compared in Figure 4.4, for different mashing conditions [64].

It is important to highlight that, for linking the mashing model and the fermentation model, sugars concentration units are considered. Despite fermentable sugars concentration is measured in Brix in the mashing model, and in °Plato in the fermentation model, any conversion is needed. At the end, as it has been explained, Brix degrees in the mashing model are measuring fermentable sugars in percentage by mass, while degrees Plato also make so. The kinetic model is going to be described hereafter.

The starch concentration evolution follows the equation 4.9, where $C_A(t)$ is the starch concentration along time (t) and k is a kinetic constant, such that:

$$\frac{dC_A(t)}{dt} = -k \cdot C_A(t) \quad (4.9)$$

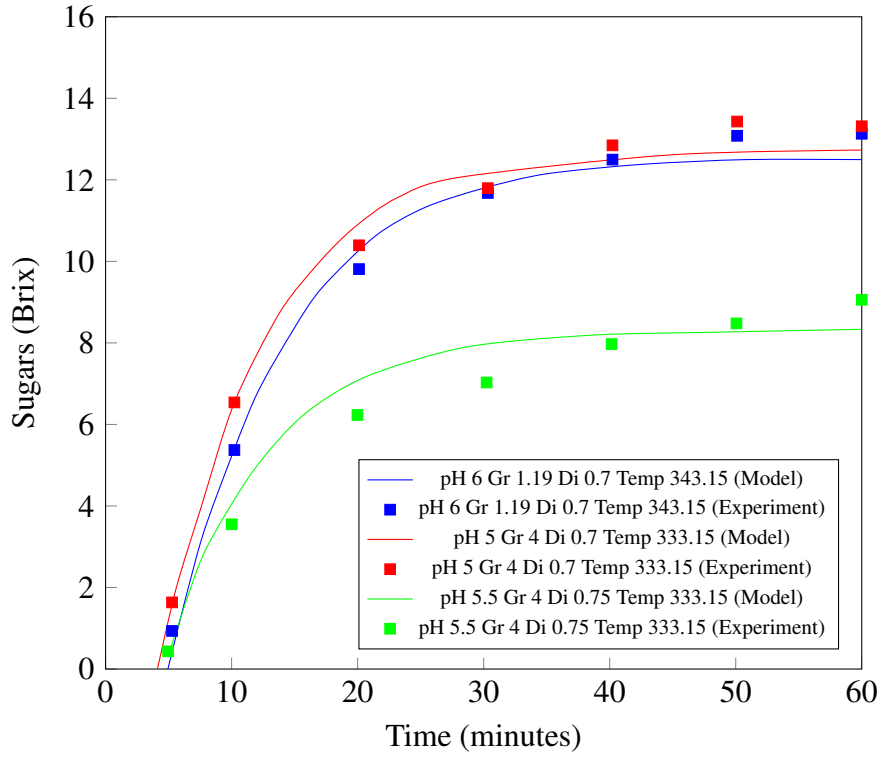


Fig. 4.4. Comparison between batch mashing model and experimental results. Several conditions. [64].

By integrating equation 4.9, it can be stated that:

$$C_A(t) = C_{A_0} \cdot e^{-k \cdot t} \quad (4.10)$$

It can be highlighted that k is an Arrhenius constant, following:

$$k = k_0 \exp\left(\frac{-E_a}{R \cdot T}\right) \quad (4.11)$$

In equation 4.11 k_0 is a pre-exponential term, E_a the activation energy, R the universal constant of ideal gases and T the temperature.

On the other hand, the evolution of the products, that is, the fermentable sugars, follows equation 4.12. In it, $C_g(t)$ is the sugar concentration along time and q is a stoichiometric coefficient.

$$C_g(t) = q \cdot (C_{A_0} - C_A) \quad (4.12)$$

By combining equation 4.10 and 4.12 we obtain equation 4.13.

$$C_g(t) = q \cdot C_{A_0} \cdot (1 - e^{-k \cdot t}) \quad (4.13)$$

That way, $q \cdot C_{A_0}$ is defined as a non time varying term linearly dependent on temperature, pH, malt grain size and dilution, following equation 4.14. These variables are named, respectively, T (K), pH , Gr (mm), and Di (%). A , B , C , D and E are empirical parameters.

$$q \cdot C_{A_0} = A \cdot T + B \cdot pH + C \cdot Gr + D \cdot Di + E \quad (4.14)$$

To conclude, the proposed model of sugars formation along is built considering the previous equations, resulting the equation 4.15.

$$C_g(t) = (A \cdot T + B \cdot pH + C \cdot Gr + D \cdot Di + E) \cdot (1 - e^{-k \cdot t}) \quad (4.15)$$

4.3.1. Mashing model. From batch to continuous operation

Before adapting the batch mashing model to a continuous one, some modifications in the way that equations are proposed by the authors are performed. The reason is for adapt them such that they will be suitable for a state-space representation. The purpose is to isolate the evolution rate of the sugars and starch concentration, that is, the time derivatives of them. Thus, the two main equation with the sugar and starch evolution rate isolated are equation 4.9, previously exposed, and equation 4.16.

$$\frac{dC_g(t)}{dt} = \frac{k}{C_{A_0}} \cdot (A \cdot T + B \cdot pH + C \cdot Gr + D \cdot Di + E) \cdot C_A \quad (4.16)$$

The way of obtaining equation 4.16 is explained hereafter. First, equation 4.12 has been derived, leading to equation 4.17, as C_{A_0} is a constant.

$$\frac{dC_g(t)}{dt} = -q \cdot \frac{dC_A(t)}{dt} \quad (4.17)$$

Then, by combining equation 4.17 with equation 4.9, it is obtained equation 4.18.

$$\frac{dC_g(t)}{dt} = q \cdot k \cdot C_A \quad (4.18)$$

And, finally, equation 4.16 is obtained by combining equations 4.18 and 4.14. In this way, the derivatives are isolated, and parameters influence (temperature, pH, grain size and dilution) is kept.

Now, as it has been done with the fermentation model, the mashing model is going to be adapted to a cascade of tanks following the same criteria. This criteria leads to the following expressions:

$$\frac{dC_{A_i}(t)}{dt} = D_{m_i} \cdot (C_{A_{i-1}}(t) - C_{A_i}(t)) - k \cdot C_{A_i}(t) \quad (4.19)$$

$$\frac{dC_{g_i}(t)}{dt} = D_{m_i} \cdot (C_{g_{i-1}}(t) - C_{g_i}(t)) + \frac{k_i}{C_{A_{initial}}} \cdot (A \cdot T_i + B \cdot pH + C \cdot Gr + D \cdot Di + E) \cdot C_{A_i} \quad (4.20)$$

On them, D_m is the dilution rate in the mashing process, similar as in the fermentation one. It has to be pointed that the naming of C_{A_0} in Equation 4.16 has been changed now to $C_{A_{initial}}$, to differentiate it from the incoming flow going into the system. Furthermore, $C_{A_{i-1}}$ for the tank 1, that is, C_{A_0} represents the initial starch concentration that is being introduced into the first tank, so it is also stated that $C_{A_0} = C_{A_{initial}}$. On the other hand, $C_{g_0} = 0$, as not any sugar content is being initially added. All other constant parameters, except temperature, remain the same for all the tanks, as it is assumed to be the same mashing process divided in four steps. Finally, it is pointed that k_0 is given in min^{-1} , so it is transformed into d^{-1} to match fermentation system time units.

5. STATE-SPACE FORM AND LINEARIZED MODEL

5.1. Dynamic models and differential equations

A mathematical model of a dynamic system has been built. On it, the relationships are established by differential equations. The components of a model are described hereafter in Table 5.1 to allow further explanations. Overall, models are made of quantities classified in variables and constants, depending if they vary with time or not, respectively.

Type	Sub-type	Definition	Notation
Constants	System parameter	Constant given by the system	
	Design Parameter	Constant that can be varied to modify the system properties	
Variables	Output	Variable which has the interest	y
	Input	External signal that can be varied along time	u
	Internal variable	Variable of the system that is neither an external signal nor an output	x
	Disturbance	External signal that can not be influenced	w

Table 5.1. COMPONENTS OF A MODEL. COMPILED FROM [66].

One way of describing these differential equations is to represent them as a first-order differential equations systems introducing internal variables [66]. Hereafter, differential equations systems representation is described according to [66]. In a vector notation, these variables can be denoted as:

$$x(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix} \quad (5.1)$$

A first-order differential equations systems can be expressed as in Equation 5.2, where $\dot{x}$ represents differentiation with respect to time and $f(x, u)$ is a vector function with n components. The functions that integrate this vector function have $n + m$ variables, which are the components of x and u vectors.

$$\dot{x}(t) = f(x(t), u(t)) \quad (5.2)$$

As a dynamic system output is all earlier values dependent, the output of the differential systems of equations expression is denoted in Equation 5.3. That is, it is calculated

from the internal variables and inputs.

$$y(t) = h(x(t), u(t)) \quad (5.3)$$

5.2. State-space representation

State-space modeling method is a vector-matrix approach of represent dynamic models, whose main advantage is the ability to model a large number degrees of freedom systems Lobontiu and it is used as a standard model for dynamic systems [66]. Moreover, they describe the state of system along time, in contrast with other techniques as the transfer function which only relates output and input; facilitate the solution of control problems and are easy to be simulated by computers.

As the output of a dynamic system depends on all earlier variables, it is not enough to know the value of the inputs u , but also the information of the internal variables x [66]. That is, the state of the system is needed, which is "the information that has to be stored and updated at the simulation" so that the output can be calculated [66]. This is from where the state-space concept comes from.

State-space models can be linear or nonlinear. Finding a solution for a nonlinear dynamic system can be problematic, so linearization can be done around an equilibrium point [67]. Also, it facilitates control techniques. Therefore, the physical model previously built is linearized in Sections 5.4 and 5.5. A linear state-space model is represented with Equations 5.4 and 5.5. On them, A is the state matrix, B the input matrix, C the output matrix, and D the direct transmission matrix [67]. Their size are $n \times n$, $n \times m$, $p \times n$ and $p \times m$ respectively [66], [67]. If these matrices are time independent, the model is linear and time invariant [66].

$$\dot{x}(t) = Ax + Bu \quad (5.4)$$

$$y(t) = Cx + Du \quad (5.5)$$

Finally, the stability concept is introduced hereafter. A system is stable when, after exciting its input, it returns to the steady-state operational point. Stability can be analyzed in state-space models by computing the eigenvalues of the matrix A , such that, if all of them are negative, the system is stable.

5.3. Linearization

Systems can be linearized around the equilibrium point if small deviations are considered [66], [67], consequently obtaining Equations 5.6 and 5.7. On them, A , B , C and D are

the Jacobian of f and h around the equilibrium point, that is, they are partial derivative matrices [66]. That means that they are obtained by taking the first order term of Taylor expansions around the equilibrium point [67].

$$\Delta \dot{x}(t) = A\Delta x + B\Delta u \quad (5.6)$$

$$\Delta y(t) = C\Delta x + D\Delta u \quad (5.7)$$

However, it is often neglected the variations in variables when representing them after linearization. Therefore, after linearizing the model in the following sections, this notation is not followed neither for simplicity, but it has to be highlighted that, after linearization, variables represent small increments.

Additionally, some limitations of linearized models have to be exposed. First, it can be performed only to study local properties around a stationary solution, which is particularly interesting if it is desired to remain the system around this point [66]. This is the aim of this project: to achieve a steady beer production. Secondly, it is difficult to estimate how accurate is the linearized approximation [66], so they might be difference between the linearized and the real model. Thus, linearized solutions "should be treated with a considerable amount of caution", and to complement them with an original model simulation is suggested [66].

5.4. Linearization and state-space model of mashing

All the equations of the mashing model are exposed hereafter (Equations 5.8-5.12). They can be assumed to be linear, as temperature (T) is assumed constant (which implies that k is constant too), as well as the pH , Gr , and Di and of course the A , B , C , D and E empirical parameters. Also it is the case for the dilution rate D_m , as the model is performed in the operational point, where the dilution rate is constant. Consequently, linearization is not needed for the mashing model.

$$\frac{dC_{A_1}(t)}{dt} = D_{m_i} \cdot (C_{A_0}(t) - C_{A_1}(t)) - k_1 \cdot C_{A_1}(t) \quad (5.8)$$

$$\frac{dC_{A_2}(t)}{dt} = D_{m_i} \cdot (C_{A_1}(t) - C_{A_2}(t)) - k_2 \cdot C_{A_2}(t) \quad (5.9)$$

$$\frac{dC_{A_3}(t)}{dt} = D_{m_i} \cdot (C_{A_2}(t) - C_{A_3}(t)) - k_3 \cdot C_{A_3}(t) \quad (5.10)$$

$$\frac{dC_{A_4}(t)}{dt} = D_{m_i} \cdot (C_{A_3}(t) - C_{A_4}(t)) - k_4 \cdot C_{A_4}(t) \quad (5.11)$$

$$\frac{dC_{g1}(t)}{dt} = D_{m_i} \cdot (C_{g0}(t) - C_{g1}(t)) + \frac{k_1}{C_{A_{initial}}} \cdot (A \cdot T_1 + B \cdot pH + C \cdot Gr + D \cdot Di + E) \cdot C_{A_1} \quad (5.12)$$

$$\frac{dC_{g2}(t)}{dt} = D_{m_i} \cdot (C_{g1}(t) - C_{g2}(t)) + \frac{k_2}{C_{A_{initial}}} \cdot (A \cdot T_2 + B \cdot pH + C \cdot Gr + D \cdot Di + E) \cdot C_{A_2} \quad (5.13)$$

$$\frac{dC_{g3}(t)}{dt} = D_{m_i} \cdot (C_{g2}(t) - C_{g3}(t)) + \frac{k_3}{C_{A_{initial}}} \cdot (A \cdot T_3 + B \cdot pH + C \cdot Gr + D \cdot Di + E) \cdot C_{A_3} \quad (5.14)$$

$$\frac{dC_{g4}(t)}{dt} = D_{m_i} \cdot (C_{g3}(t) - C_{g4}(t)) + \frac{k_4}{C_{A_{initial}}} \cdot (A \cdot T_4 + B \cdot pH + C \cdot Gr + D \cdot Di + E) \cdot C_{A_4} \quad (5.15)$$

Thereby, once the equations are known to be linear, the state-space model is built. According to [68], equations are often unstructured, and they may be organized. However, in this project, mashing equations have been already organized and oriented to build a state-space model with them, as it has been made in Section 4.3. The original batch mashing model was giving the evolution of starch concentration along time, and it has been converted to an expression that describes the derivative of starch concentration along time, that is, into a differential equation, with the purpose of making it suitable for a state-space model.

The steps to build an state-space, according to [68], are to choose the state variables, to express their time derivative in terms of other states variables and inputs, and to express the outputs as functions of state variables and inputs.

Nevertheless, in this project, these steps have been intrinsically followed previously while building the physical model. That is the reason of why the original batch model, which described sugar concentration evolution, instead of sugar concentration derivative evolution, was adapted to fulfill this requirement. Therefore, the state variables, inputs, and outputs, were intrinsically easily chosen previously. Consequently, state variables, inputs, and outputs of the continuous mashing model are summarized in Table 5.2.

It is obvious to understand why C_{A_0} and C_{g4} are chosen as input and output respectively. The system is fed with an input flow of hot water and grist with a concentration C_{A_0} . The output is C_{g4} as the outlet sugar concentration will be working as the fermentation input.

Before building the state space system matrices, the sugar derivative expression is

Continuous Mashing State-Space Model Components	
State Variables	$C_{A_1}, C_{A_2}, C_{A_3}, C_{A_4},$ $C_{g_1}, C_{g_2}, C_{g_3}, C_{g_4}$
Inputs	C_{A_0}
Outputs	C_{g_4}

Table 5.2. STATE VARIABLES, INPUTS AND OUTPUTS OF THE CONTINUOUS MASHING STATE SPACE MODEL.

simplified for typing space purposes, as it is expressed in Equation 5.16.

$$W_i = D_{m_i} \cdot \frac{k_i}{C_{A_{initial}}} \cdot (A \cdot T_i + B \cdot pH + C \cdot Gr + D \cdot Di + E) \quad (5.16)$$

Therefore, the state space matrices are shown in Equations 5.17-5.23. Finally, it is reminded that, as it has been exposed in Section 4.3.1, $C_{g_0} = 0$, as not any sugar content is being initially added. The eigenvalues of A_m are all negative, so the system is stable.

$$x_m(t) = \begin{bmatrix} C_{A_1}(t) \\ C_{A_2}(t) \\ C_{A_3}(t) \\ C_{A_4}(t) \\ C_{g_1}(t) \\ C_{g_2}(t) \\ C_{g_3}(t) \\ C_{g_4}(t) \end{bmatrix} \quad (5.17)$$

$$\dot{x}_m(t) = \begin{bmatrix} \dot{C}_{A_1}(t) \\ \dot{C}_{A_2}(t) \\ \dot{C}_{A_3}(t) \\ \dot{C}_{A_4}(t) \\ \dot{C}_{g_1}(t) \\ \dot{C}_{g_2}(t) \\ \dot{C}_{g_3}(t) \\ \dot{C}_{g_4}(t) \end{bmatrix} \quad (5.18)$$

$$B_m = \begin{bmatrix} D_{m_1} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (5.19)$$

$$A_m = \begin{bmatrix} -D_{m_1} - k_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ D_{m_2} & -D_{m_2} - k_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & D_{m_3} & -D_{m_3} - k_3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & D_{m_4} & -D_{m_4} - k_4 & 0 & 0 & 0 & 0 \\ W_1 & 0 & 0 & 0 & -D_{m_1} & 0 & 0 & 0 \\ 0 & W_2 & 0 & 0 & D_{m_2} & -D_{m_2} & 0 & 0 \\ 0 & 0 & W_3 & 0 & 0 & D_{m_3} & -D_{m_3} & 0 \\ 0 & 0 & 0 & W_4 & 0 & 0 & D_{m_4} & -D_{m_4} \end{bmatrix} \quad (5.20)$$

$$C_m = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (5.21)$$

$$D_m = 0 \quad (5.22)$$

$$u_m = C_{A_0} \quad (5.23)$$

5.5. Linearization and state-space model of fermentation

In the case of the fermentation model, unlike the mashing one, it is non-linear and it has to be linearized. The complete equations of the models are exposed below (Equations 5.24-5.35).

$$\frac{dS_1(t)}{dt} = D_{f_1} \cdot (S_0 - S_1) + \left[-\frac{1}{Y_{X/S}} \cdot \left(\mu_{max} \cdot \frac{S_1(t)}{K_{sx} + S_1(t)} \cdot X(t) \right) - \frac{1}{Y_{P/S}} \cdot \left(q_{pmax} \cdot \frac{S_1(t)}{K_{sp} + S_1(t)} \cdot X(t) \right) \right] \quad (5.24)$$

$$\frac{dS_2(t)}{dt} = D_{f_2} \cdot (S_1 - S_2) + \left[-\frac{1}{Y_{X/S}} \cdot \left(\mu_{max} \cdot \frac{S_2(t)}{K_{sx} + S_2(t)} \cdot X(t) \right) - \frac{1}{Y_{P/S}} \cdot \left(q_{pmax} \cdot \frac{S_2(t)}{K_{sp} + S_2(t)} \cdot X(t) \right) \right] \quad (5.25)$$

$$\frac{dS_3(t)}{dt} = D_{f_3} \cdot (S_2 - S_3) + \left[-\frac{1}{Y_{X/S}} \cdot \left(\mu_{max} \cdot \frac{S_3(t)}{K_{sx} + S_3(t)} \cdot X(t) \right) - \frac{1}{Y_{P/S}} \cdot \left(q_{pmax} \cdot \frac{S_3(t)}{K_{sp} + S_3(t)} \cdot X(t) \right) \right] \quad (5.26)$$

$$\frac{dS_4(t)}{dt} = D_{f_4} \cdot (S_3 - S_4) + \left[-\frac{1}{Y_{X/S}} \cdot \left(\mu_{max} \cdot \frac{S_4(t)}{K_{sx} + S_4(t)} \cdot X(t) \right) - \frac{1}{Y_{P/S}} \cdot \left(q_{pmax} \cdot \frac{S_4(t)}{K_{sp} + S_4(t)} \cdot X(t) \right) \right] \quad (5.27)$$

$$\frac{dX_1(t)}{dt} = D_{f_1} \cdot (X_0 - X_1) + \mu_{max} \cdot \frac{S_1(t)}{K_{sx} + S_1(t)} \cdot X(t) \quad (5.28)$$

$$\frac{dX_2(t)}{dt} = D_{f_2} \cdot (X_1 - X_2) + \mu_{max} \cdot \frac{S_2(t)}{K_{sx} + S_2(t)} \cdot X(t) \quad (5.29)$$

$$\frac{dX_3(t)}{dt} = D_{f_3} \cdot (X_2 - X_3) + \mu_{max} \cdot \frac{S_3(t)}{K_{sx} + S_3(t)} \cdot X(t) \quad (5.30)$$

$$\frac{dX_4(t)}{dt} = D_{f_4} \cdot (X_3 - X_4) + \mu_{max} \cdot \frac{S_4(t)}{K_{sx} + S_4(t)} \cdot X(t) \quad (5.31)$$

$$\frac{dP_1(t)}{dt} = D_{f_1} \cdot (P_0 - P_1) + q_{pmax} \cdot \frac{S_1(t)}{K_{sp} + S_1(t)} \cdot X(t) \quad (5.32)$$

$$\frac{dP_2(t)}{dt} = D_{f_2} \cdot (P_1 - P_2) + q_{pmax} \cdot \frac{S_2(t)}{K_{sp} + S_2(t)} \cdot X(t) \quad (5.33)$$

$$\frac{dP_3(t)}{dt} = D_{f_3} \cdot (P_2 - P_3) + q_{pmax} \cdot \frac{S_3(t)}{K_{sp} + S_3(t)} \cdot X(t) \quad (5.34)$$

$$\frac{dP_4(t)}{dt} = D_{f_4} \cdot (P_3 - P_4) + q_{pmax} \cdot \frac{S_4(t)}{K_{sp} + S_4(t)} \cdot X(t) \quad (5.35)$$

Several non-linear terms can be identified. As it is explained in Section 5.3, the non-linear terms are linearized using Taylor expansions around the equilibrium point. Hereafter, the terms are identified and linearized.

$$f_1 = \frac{S_i(t)}{K_{sp} + S_i(t)} \cdot X_i(t) \quad (5.36)$$

$$f_1 \approx \left. \frac{\partial f_1}{\partial S_i} \right|_{S_i^{eq}, X_i^{eq}} \cdot S_i(t) + \left. \frac{\partial f_1}{\partial X_i} \right|_{S_i^{eq}, X_i^{eq}} \cdot X_i(t) \quad (5.37)$$

$$f_1 \approx \frac{(K_{sp} + S_i^{eq}) \cdot X_i^{eq} - S_i^{eq} \cdot X_i^{eq}}{(K_{sp} + S_i^{eq})^2} \cdot S_i(t) + \frac{S_i^{eq}}{K_{sp} + S_i^{eq}} \cdot X_i(t) \quad (5.38)$$

$$f_2 = \frac{S_i(t)}{K_{sx} + S_i(t)} \cdot X_i(t) \quad (5.39)$$

$$f_2 \approx \left. \frac{\partial f_1}{\partial S_i} \right|_{S_i^{eq}, X_i^{eq}} \cdot S_i(t) + \left. \frac{\partial f_1}{\partial X_i} \right|_{S_i^{eq}, X_i^{eq}} \cdot X_i(t) \quad (5.40)$$

$$f_2 \approx \frac{(K_{sx} + S_i^{eq}) \cdot X_i^{eq} - S_i^{eq} \cdot X_i^{eq}}{(K_{sx} + S_i^{eq})^2} \cdot S_i(t) + \frac{S_i^{eq}}{K_{sx} + S_i^{eq}} \cdot X_i(t) \quad (5.41)$$

Therefore, the linearized equations are hereafter expressed (Equations 5.42-5.44).

$$\begin{aligned} \frac{dS_i(t)}{dt} = D_{f_i} \cdot (S_{i-1} - S_i) &+ \left[-\frac{1}{Y_{X/S}} \cdot \mu_{max} \cdot \left(\frac{(K_{sx} + S_i^{eq}) \cdot X_i^{eq} - S_i^{eq} \cdot X_i^{eq}}{(K_{sx} + S_i^{eq})^2} \cdot S_i(t) + \frac{S_i^{eq}}{K_{sx} + S_i^{eq}} \cdot X_i(t) \right) \right. \\ &\left. - \frac{1}{Y_{P/S}} \cdot q_{pmax} \cdot \left(\frac{(K_{sp} + S_i^{eq}) \cdot X_i^{eq} - S_i^{eq} \cdot X_i^{eq}}{(K_{sp} + S_i^{eq})^2} \cdot S_i(t) + \frac{S_i^{eq}}{K_{sp} + S_i^{eq}} \cdot X_i(t) \right) \right] \end{aligned} \quad (5.42)$$

$$\frac{dX_i(t)}{dt} = D_{f_i} \cdot (X_{i-1} - X_i) + \mu_{max} \cdot \left(\frac{(K_{sx} + S_i^{eq}) \cdot X_i^{eq} - S_i^{eq} \cdot X_i^{eq}}{(K_{sx} + S_i^{eq})^2} \cdot S_i(t) + \frac{S_i^{eq}}{K_{sx} + S_i^{eq}} \cdot X_i(t) \right) \quad (5.43)$$

$$\frac{dP_i(t)}{dt} = D_{f_i} \cdot (P_{i-1} - P_i) + q_{pmax} \cdot \left(\frac{(K_{sp} + S_i^{eq}) \cdot X_i^{eq} - S_i^{eq} \cdot X_i^{eq}}{(K_{sp} + S_i^{eq})^2} \cdot S_i(t) + \frac{S_i^{eq}}{K_{sp} + S_i^{eq}} \cdot X_i(t) \right) \quad (5.44)$$

The state-space model is built as it has been done in Section 5.4. Equations are here also already organized to build a state-space model, with the fermentable sugars, yeast and alcohol time derivatives already isolated. State variables, inputs, and outputs are chosen and described in Table 5.3.

Continuous Mashing State-Space Model Components	
State Variables	$S_1, S_2, S_3, S_4,$ $X_1, X_2, X_3, X_4,$ P_1, P_2, P_3, P_4
Inputs	S_0
Outputs	S_4, X_4, P_4

Table 5.3. STATE VARIABLES, INPUTS AND OUTPUTS OF THE CONTINUOUS MASHING STATE SPACE MODEL.

Simillarly to the mashing state-space model, in this case S_0 is chosen as the input, as the fermentation cascade system is fed with wort. Thus, S_0 is equivalent to the outcome wort of the mashing step, C_{g4} . Regarding the selection of the outputs, the three variables are selected to monitor the final beer composition.

Again, before building the state-space system matrices, some expressions are simplified for typing purposes. Such simplifications are shown in Equations 5.45-5.48.

$$Q_i = \mu_{max} \cdot \frac{(K_{sx} + S_i^{eq}) \cdot X_i^{eq} - S_i^{eq} \cdot X_i^{eq}}{(K_{sx} + S_i^{eq})^2} \quad (5.45) \quad T_i = \mu_{max} \cdot \frac{S_i^{eq}}{K_{sx} + S_i^{eq}} \quad (5.46)$$

$$U_i = q_{pmax} \cdot \frac{(K_{sp} + S_i^{eq}) \cdot X_i^{eq} - S_i^{eq} \cdot X_i^{eq}}{(K_{sp} + S_i^{eq})^2} \quad (5.47) \quad V_i = q_{pmax} \cdot \frac{S_i^{eq}}{K_{sp} + S_i^{eq}} \quad (5.48)$$

The state space matrices are shown in Equations 5.49-5.55. Finally, it is reminded that, as it has been explained in Section 4.2.1, the supply of yeast X_0 and alcohol P_0 is equal to zero, and the incoming sugar concentration, S_0 , is equivalent to the output sugar concentration of the mashing state C_{g4} . Regarding stability, the eigenvalues of A_f are all negative.

$$x_f(t) = \begin{bmatrix} S_1(t) \\ S_2(t) \\ S_3(t) \\ S_4(t) \\ X_1(t) \\ X_2(t) \\ X_3(t) \\ X_4(t) \\ P_1(t) \\ P_2(t) \\ P_3(t) \\ P_4(t) \end{bmatrix} \quad (5.49) \quad \dot{x}_f(t) = \begin{bmatrix} \dot{S}_1(t) \\ \dot{S}_2(t) \\ \dot{S}_3(t) \\ \dot{S}_4(t) \\ \dot{X}_1(t) \\ \dot{X}_2(t) \\ \dot{X}_3(t) \\ \dot{X}_4(t) \\ \dot{P}_1(t) \\ \dot{P}_2(t) \\ \dot{P}_3(t) \\ \dot{P}_4(t) \end{bmatrix} \quad (5.50) \quad B_f = \begin{bmatrix} D_{f1} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (5.51)$$

$$A_f = \begin{bmatrix} -D_{f1} - (Q_1 + U_1)/Y_{X/S} & 0 & 0 & 0 & -(T_1 + V_1)/Y_{P/S} & 0 & 0 & 0 & 0 & 0 & 0 \\ D_{f2} & -D_{f2} - (Q_2 + U_2)/Y_{X/S} & 0 & 0 & 0 & -(T_2 + V_2)/Y_{P/S} & 0 & 0 & 0 & 0 & 0 \\ 0 & D_{f3} & -D_{f3} - (Q_3 + U_3)/Y_{X/S} & 0 & 0 & 0 & -(T_3 + V_3)/Y_{P/S} & 0 & 0 & 0 & 0 \\ 0 & 0 & D_{f4} & -D_{f4} - (Q_4 + U_4)/Y_{X/S} & 0 & 0 & 0 & -(T_4 + V_4)/Y_{P/S} & 0 & 0 & 0 \\ Q_1 & 0 & 0 & 0 & -T_1 - D_{f1} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & Q_2 & 0 & 0 & D_{f2} & -T_2 - D_{f2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & Q_3 & 0 & 0 & D_{f3} & -T_3 - D_{f3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_4 & 0 & 0 & D_{f4} & -T_4 - D_{f4} & 0 & 0 & 0 \\ U_1 & 0 & 0 & 0 & V_1 & 0 & 0 & 0 & -D_{f1} & 0 & 0 \\ 0 & U_2 & 0 & 0 & 0 & V_2 & 0 & 0 & D_{f2} & -D_{f2} & 0 \\ 0 & 0 & U_3 & 0 & 0 & 0 & V_3 & 0 & 0 & D_{f3} & -D_{f3} \\ 0 & 0 & 0 & U_4 & 0 & 0 & 0 & V_4 & 0 & 0 & D_{f4} - D_{f4} \end{bmatrix} \quad (5.52)$$

$$C_f = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (5.53)$$

$$D_f = 0 \quad (5.54)$$

$$u_f = S_0 \quad (5.55)$$

5.6. Concatenated state-space models

Once both steps, mashing and fermentation, have been modelled, they are joint to be simulated together, so a simulation of the performance of the whole process is computed. Making different state-space models for each subsystem is often advantageous, especially for large systems, as the model becomes modularized and the behaviour of the model can be checked independently [69]. Also, it gives the possibility to replace independently the subsystems if it is desired to consider another subsystem model.

In the case of this project, the aim is to combine the mashing and fermentation models in series. This can be easily made with algebra and partitioned matrices or submatrices.

From [70], the state-space representation of two concatenated systems is obtained. Therefore, the state-space representation of the mashing and fermentation models concatenated follows Equations 5.56-5.57.

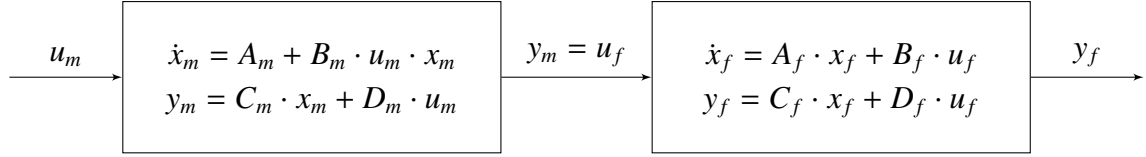


Fig. 5.1. State-space representation for concatenated systems. Adapted from [70].

$$\begin{bmatrix} \dot{x}_m \\ \dot{x}_f \end{bmatrix} = \begin{bmatrix} A_m & 0 \\ B_f \cdot C_m & A_f \end{bmatrix} \cdot \begin{bmatrix} x_m \\ x_f \end{bmatrix} + \begin{bmatrix} B_m \\ B_f \cdot D_m \end{bmatrix} \cdot u_m \quad (5.56)$$

$$y_f = \begin{bmatrix} D_f \cdot C_m & C_f \end{bmatrix} \cdot \begin{bmatrix} x_m \\ x_f \end{bmatrix} + \begin{bmatrix} D_f \cdot D_m \end{bmatrix} \cdot u_m \quad (5.57)$$

Nevertheless, these expressions are not used in this project, and are written just to show the theoretical background of the concatenated state-space systems. They would not be necessary to use as Simulink, the software that is used to perform the simulations in this project, allows to directly link the output of one state-space system to the input of another. However, still this is not used, because finally the fermentation state-space model is going to be neglected as it does not reach the same steady-state point than the non-linearized model. This is explained later in Section 6.2.

6. SIMULATION

Simulation is the next step in the procedure of modelling a system. In fact, usually models are built oriented to perform a simulation, as in the case of state-space models, explained in Chapter 5. Being simulation a cheap and safe option to experiment with a system [71], one of its common goals is to assist the design phase finding appropriate design parameters values [66]. In fact, during the design process, it is used as soon as possible, so design costs and inappropriate designed are minimized and eliminated, practice that is called "Simulation-Based Design" [72] (SBD). Consequently, Simulation-Based Design saves effort, time and resources [72].

Thus, simulation can help to design industrial processes, which have been traditionally designed for steady state operational points [69]. This is the case of this project, in which simulation is carried to give suitable design parameters in a stationary brewing operation, with a SBD approach.

Moreover, although simulation is a technique that has been carried out before the Industry 4.0 term was coined, in [73]–[75] modelling and simulation are proposed as key technologies to innovate and develop the Industry 4.0.

Finally, besides simulation strengths, some of its disadvantages are the lack of simulation engineers and their high salaries and the needed time to develop models [76], [77].

There are several software options available to perform simulations. In this project, Simulink has been chosen to perform them. Simulink is a powerful MATLAB toolbox to simulate systems. It can describe a system in several ways, and offers different input signals generation possibilities [69]. It has, of course, and excellent MATLAB integration, a huge variety of options, and the possibility to simulate linear and non-linear systems. Its graphical environment offers a good visualization and understanding and, therefore, an easy learning. Besides, it is a wide used software, with a huge amount of solved doubts on forums and a big community that can be a support for any kind of doubt.

The simulation of the system, which has been modelled for a continuous steady-state operation, is not presenting any response if any input is excited. That is, is stable, as it has been previously deduced in Section 5.4 by pointing that the eigenvalues of A_m are all negative.

Consequently, for obtaining an interesting result of the simulation, the input signal is excited, so the system dynamics can be analyzed. The constant values of the model parameters are equal to those reached in the equilibrium point, and which are described in Table 7.1.

It has to be clarified that, for visualizing all the state variables, the matrix C_m has been changed to a 8×8 identity matrix, and D_m to a 1×8 matrix with all the entries equal to zero in the case of the mashing subsystem simulation.

The system is going to be excited with a pulse signal, that is going to emulate an impulse response, and with a random disturbance signal. It is highlighted that a step response of the system is not studied as it would imply to change the operational point, and, consequently, to change the obtained beer properties, which is not desired. Prior to a simulation of the whole system, that is, the mashing and fermentation models in series, both of the two different subsystems are analyzed independently.

The following simulation results are started with the steady-state operational points that have been iteratively obtained, and that can be seen in Figure 7.1.

6.1. Continuous mashing subsystem simulation

In the case of the mashing model, as it was previously linear and a state model could be directly built (any linearization was needed), the simulation is going to be performed with the state-space Simulink block.

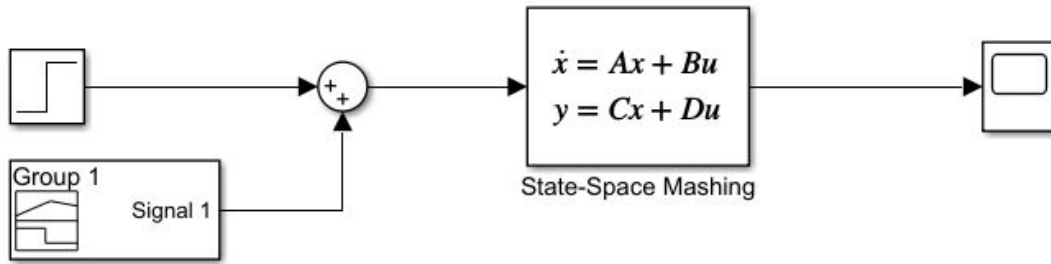


Fig. 6.1. Simulink layout of the mashing system.

The Simulink layout for this step is shown in Figure 6.1. The step signal block is actually a signal with a constant value of C_{A_0} , that is, 0.1649 w/w, according to Table 7.1. It is summed to a signal generator, that generates a pulse of amplitude 50 w/w to emulate an impulse response. The pulse signal has a value of 50 w/w when $1 \leq t \leq 1.001$, being time, t , measured in days. That is, is a pulse which lasts for 1.44 min. The pulse signal is represented in Figure 6.2.

The purpose of adding a signal that is around 300 times greater than the usual fed of starch concentration is to study how the mashing process would behave under an extreme situation. Although is not feasible to feed the mashing process with a starch concentration of 50 w/w, that is around 300 times greater than the operation starch feeding concentration, the simulation gives some interesting insights about the dynamics of the system. The output of the simulation is displayed in Figures 6.3 and 6.4. It has to be noticed that, because of scaling issues, in Figure 6.3, C_{A_4} is represented in a different axis.

From the simulation of the excitation of the system with a pulse signal, interesting insights of the system can be seen in Figures 6.3 and 6.4. First thing that can be noticed is that the system is stable, and, once the disturbance effect is finished, it comes back to the stationary point. Then, it can be observed that the sugar concentration response of a

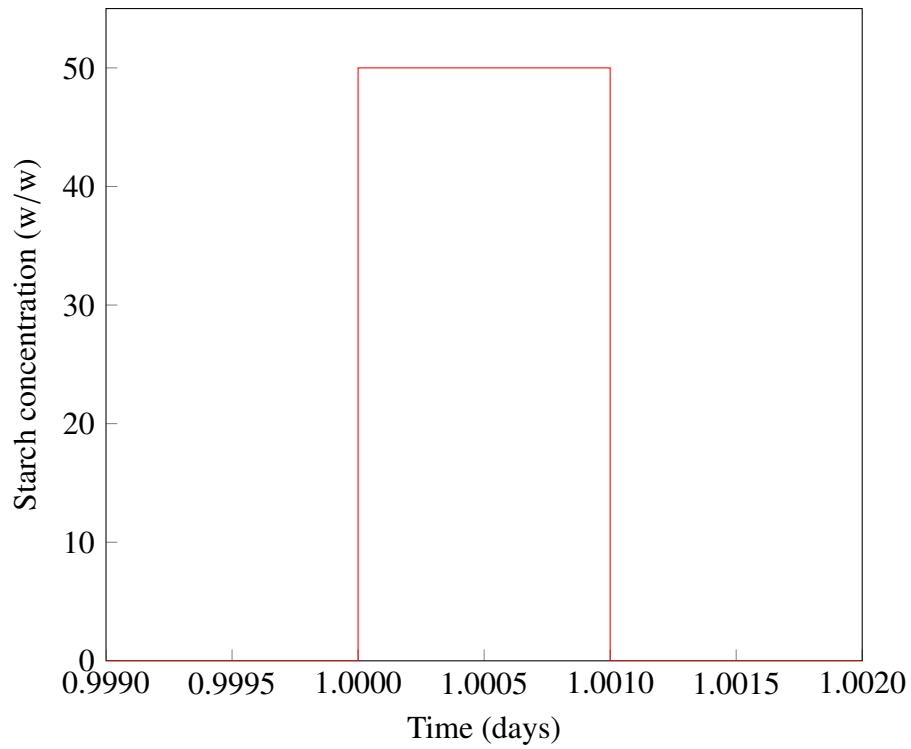


Fig. 6.2. Pulse signal for mashing simulation.

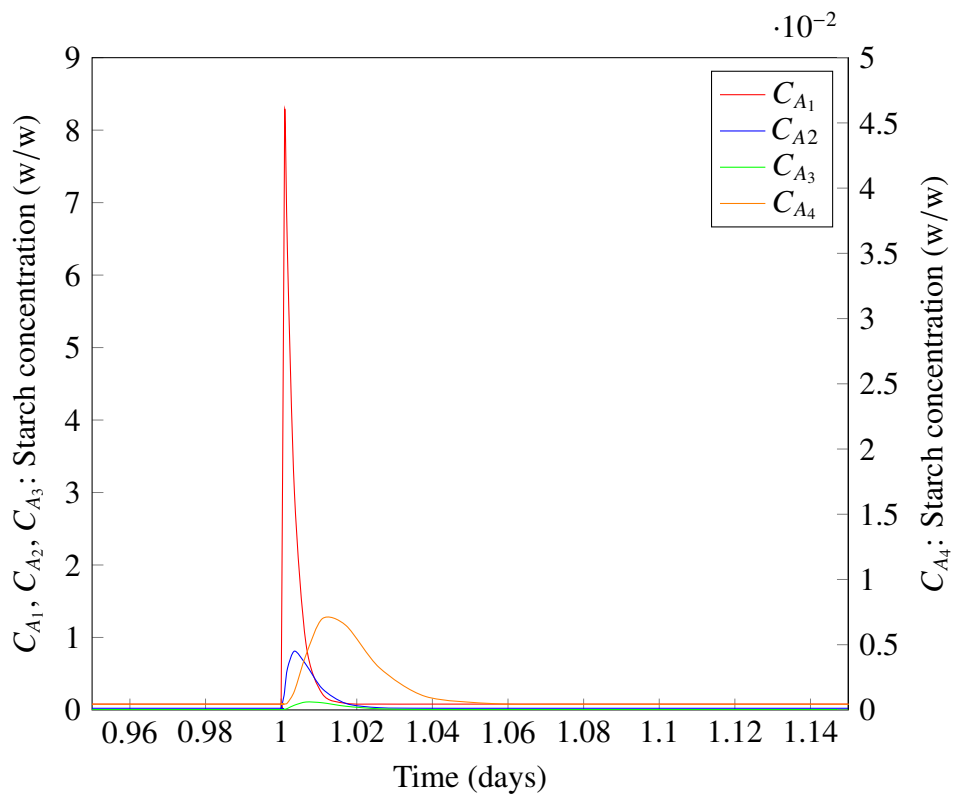


Fig. 6.3. Mashing response to a 50 w/w pulse signal. Starch concentrations.

tank with respect to the response of the starch concentration of the same tank is delayed. For example, while in the third tank the peak concentration of starch is obtained with

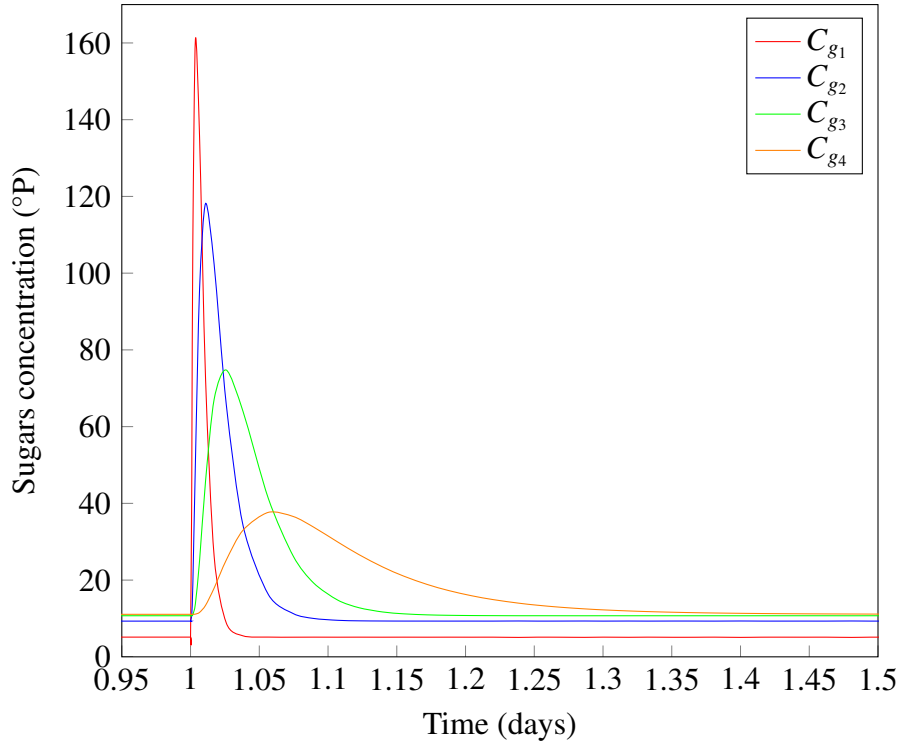


Fig. 6.4. Mashing response to a 50 w/w pulse signal. Sugars concentrations.

a difference of time of, approximately, 1.16 min. This behaviour is reasonable, as the system need some time to convert starch into sugar.

Also, it is very interesting to analyze the impact of the cascade of tanks on the system behaviour. It can be noticed that the effect of the tanks in series is not only to delay the peaks between the concentrations in one tank an the other, but also to damp the effect of them. While the first effect, that is, the delay between peaks is not important and neither desired nor undesired, the second one is a positive desired effect of the cascade configuration. It suggests that, the effect of the cascade is keeping an even output wort distribution along time. At the same time, the damping effect also makes the period of having a disturbed operational point longer: while the first tank recovers its original concentration fast, the last one takes more time to do it.

To calculate that the system is actually damping the effect of a peak input, the difference between the operational point, in percentage, is calculated for the first and the fourth tank. The first tank has an operational sugar concentration of 11.08 °P, and a maximum value of 160.86 °P, which makes a difference between the peak and the steady-state concentration of, approximately, 1351.8%. In the case of tank 4, its operation concentration is 5.139 °P, and its peak maximum value is 37.662 °P, which makes a difference of, approximately, 632.86% between them. It is reminded that this situation is not possible to happen and that the huge disturbance input has the purpose of studying the system dynamics.

Once that the mashing system dynamics have been studied with a pulse signal, a more realistic scenario is going to be simulated. The block generating the pulse signal

in Figure 6.1 is substituted by a signal generator block, to generate a disturbance signal. The disturbance signal generated has a random wave form, with an amplitude of a 10% of the non-disturbed C_{A_0} value and a frequency of 4.32 d^{-1} . The results of this simulation is displayed in Figures 6.5 and 6.6.

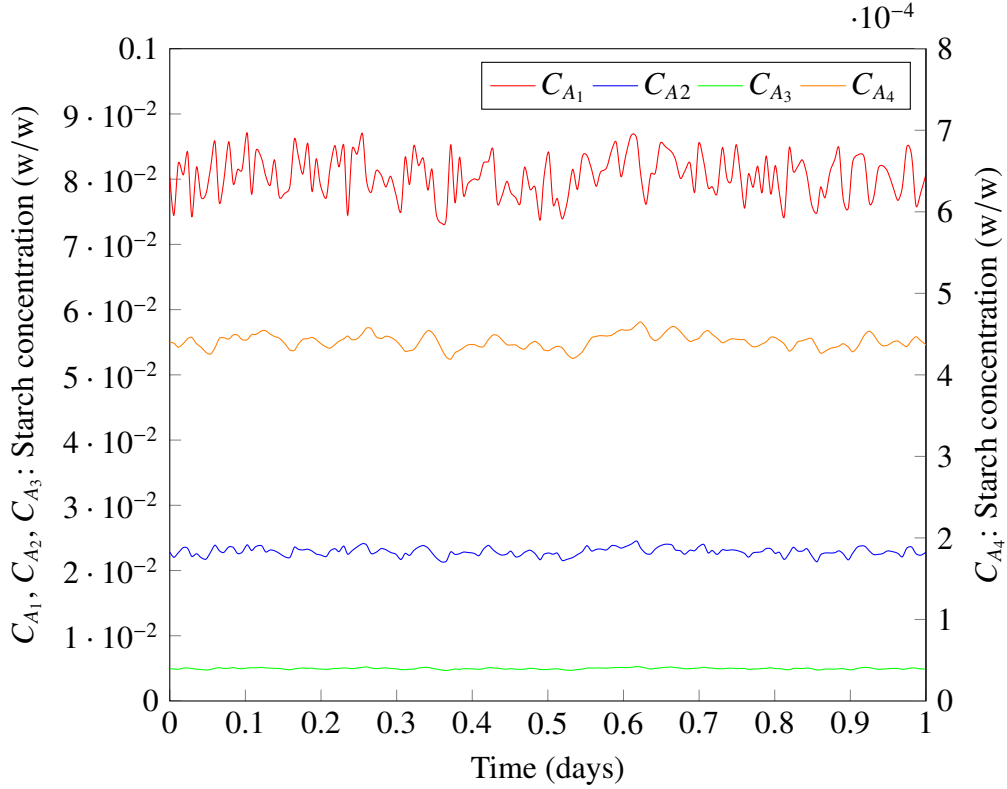


Fig. 6.5. Mashing response when adding a 10% C_{A_0} amplitude random signal. Starch concentrations.

In contrast to the pulse simulation, this new disturbance signal is more likely to be present in an actual brewery plant, as the incoming raw materials might not be completely even (a deviation of 10% in the incoming starch concentration has been introduced). A similar analysis is performed to study the damping effect of the mashing cascade reactors in a more realistic situation. Thus, the sugar concentration of the first and the fourth tanks are compared in a similar way as it has been done with the pulse signal. Visually, it can be seen in Figures 6.5 and 6.6 how the later tanks are having smoother starch and sugar concentration signals than the previous ones. Among the simulated time, the maximum difference of the sugar concentration in the first tank is of 7.23%, and in the fourth one is of 1.93%. Therefore, the damping behaviour of the cascade is also proved with a more realistic simulation, and a desired even output is obtained.

6.2. Continuous fermentation subsystem simulation

As it has been said in Section 5.3, it is suggested to complement linearized solutions with an original model simulations [66].

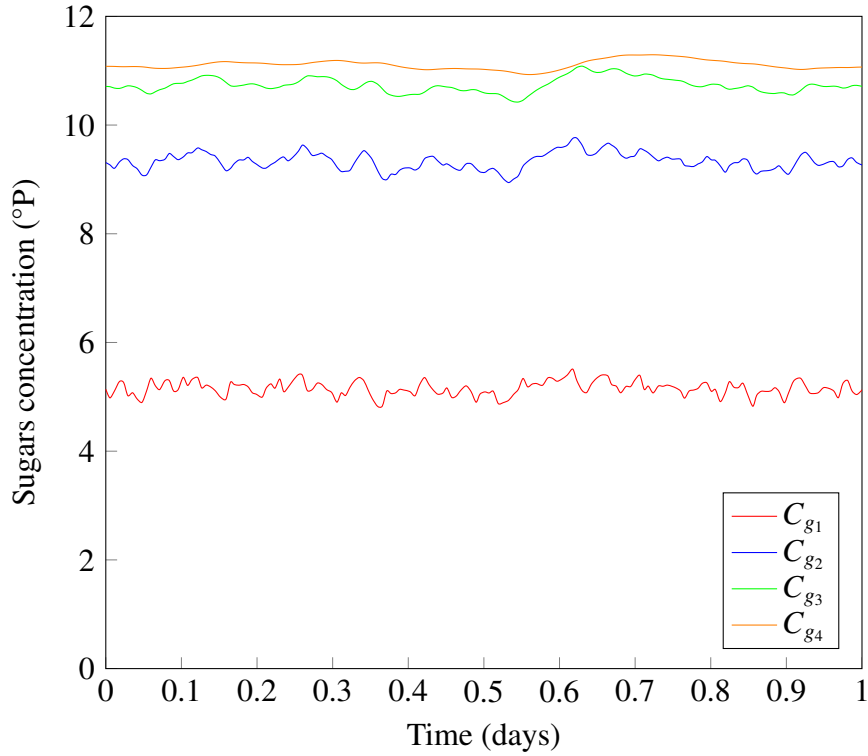


Fig. 6.6. Mashing response when adding a 10% C_{A_0} amplitude random signal. Sugars concentrations.

Thus, in this case, the non linearized continuous fermentation model has been built with Simulink blocks, as it is shown in Figures 6.14 and 6.15. After finding a desired operational point, the simulation solutions of the non linearized model and the linearized one (built with the state-space model of Section 5.5) have been compared. It has been seen that both of the systems reach a different stationary solution. For example, while the alcohol concentration stationary solution of the non linearized system is 4.364 %w/w, the alcohol concentration stationary solution of the linearized one is 1.313 %w/w. Consequently, the linearized state-space model have been neglected, and the further simulations have been done with the original model, that it, it has been used the Simulink model of Figures 6.14 and 6.15.

Once a comparison between the linearized an non linearized models have been done, the same simulation procedures of the continuous mashing stage are made for the continuous fermentation.

Now, the the step signal block is a constant value signal of $C_{g_4} = 11.08$ °P, which is the fermentable sugars concentration of the output mashing wort. However, in this case, the signal generator gives a pulse of amplitude 5000 °P when $1 \leq t \leq 1.5$, being time, t , also measured in days. The reason of why now the pulse amplitude and duration has been changed is because the original input value of the continuous fermentation system is much higher than the one of the continuous fermentation system; and because the dynamics of the fermentation are slower than the mashing dynamics. Therefore, a pulse

with different characteristics it is needed to obtain an insightful simulation output. The pulse signal for the continuous fermentation simulation is represented in Figure 6.7, and the simulation solutions to the system's response are shown in Figures 6.8, 6.9 and 6.10. It is reminded that, similarly to the mashing pulse response simulation, an input signal of these characteristics is not feasible, and it is only simulated for studying the system dynamics.

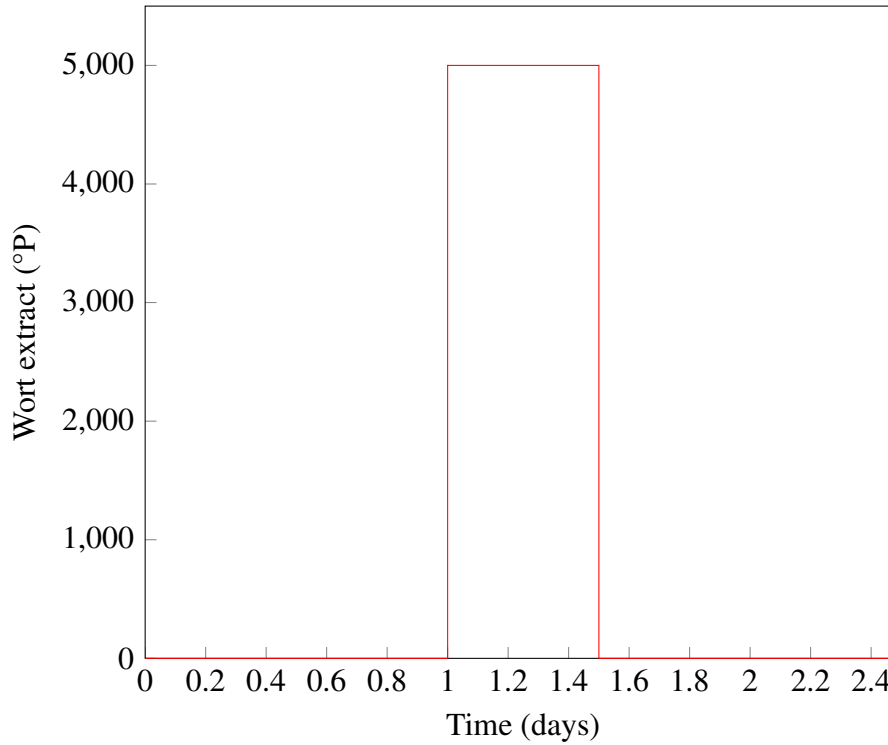


Fig. 6.7. Pulse signal for fermentation simulation.

Similarly to the mashing stage, the continuous fermentation simulations suggests the same appreciations. As it can be visually observed in Figures 6.8, 6.9 and 6.10, the system is also stable, and the impact of the cascade of tanks in the system behaviour is the same. Peaks in concentrations from different tanks are delayed, and there is a damping effect which enables to obtain an even output beer along time.

However, while in the mashing process there is only one product (the sugar concentration which depends on the starch concentration), in the fermentation process there are two. The fermentable sugars concentration is leading the yeast and alcohol peaks, as fermentable sugars are what limits the reaction. Nonetheless, it is interesting to realize that both of the yeast and alcohol peaks are happening at the same time, that is, at time $t = 391.55$ d in the case of the second tank. This might be because, as it can be seen in Equations 4.7 and 4.8, yeast and alcohol change rate depend in a similar way on sugar and yeast concentration.

Again, a more realistic scenario is considered by substituting the pulse signal by a disturbance signal. In this case, it also has a random wave form, an amplitude of a 10% of the non-disturbed C_{g4} steady-state value, and a frequency of 4.32 d^{-1} . Nevertheless,

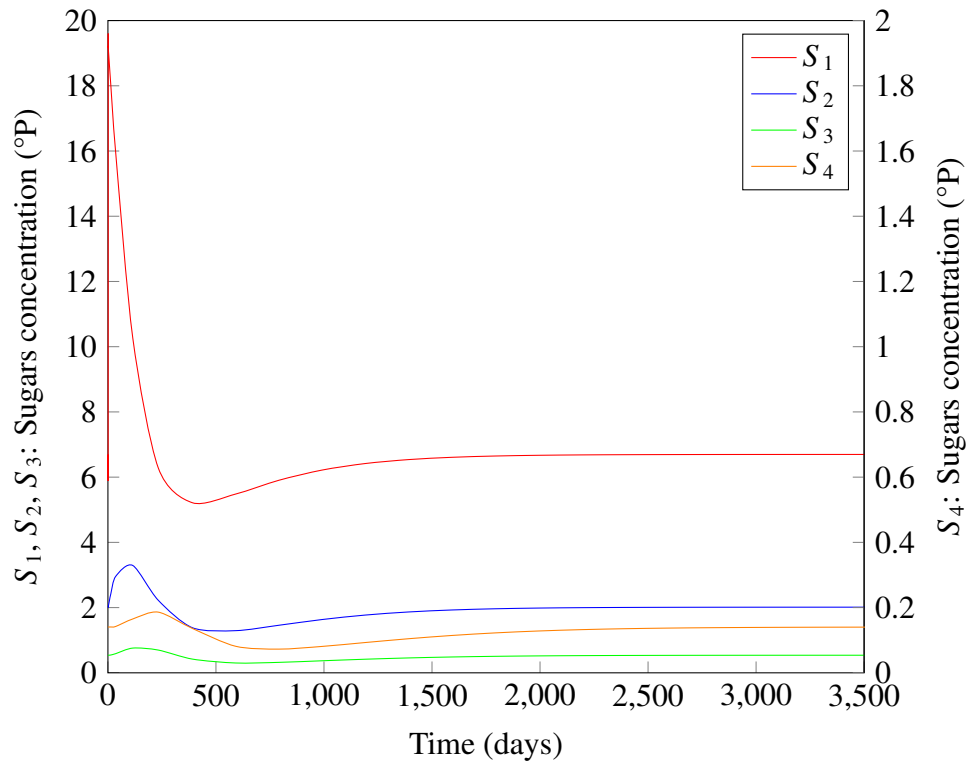


Fig. 6.8. Fermentation response to a 5000 °P pulse signal. Sugars concentrations.

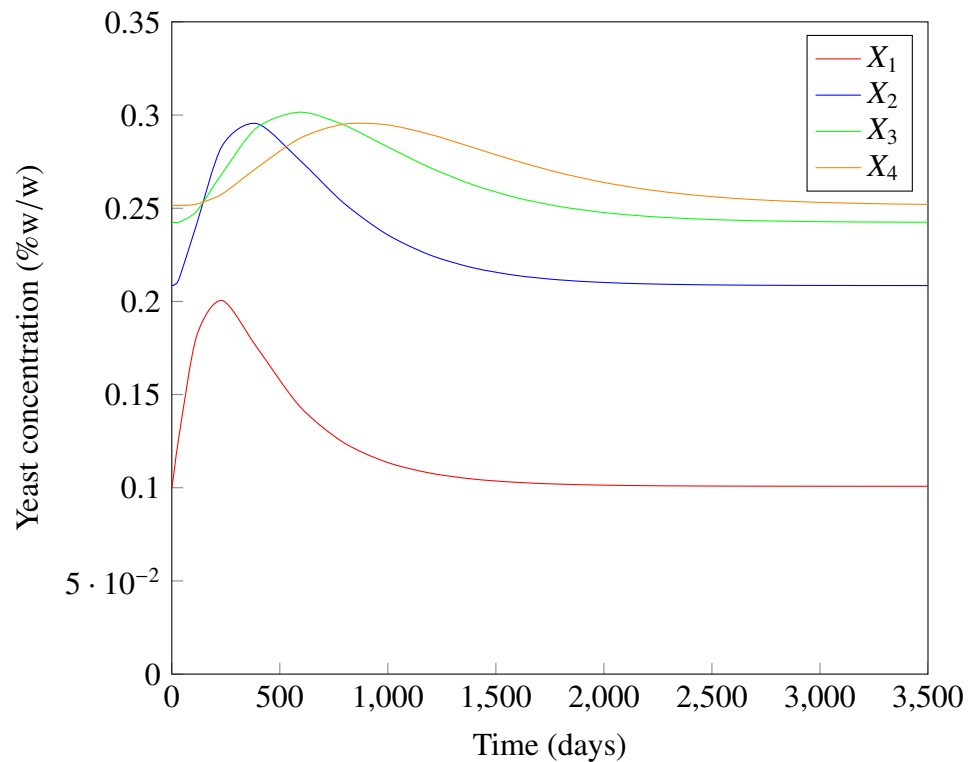


Fig. 6.9. Fermentation response to a 5000 °P pulse signal. Yeast concentrations.

this situation is still not very likely to happen, as in the mashing simulation that has been previously explained the maximum difference in the sugar concentration output is of

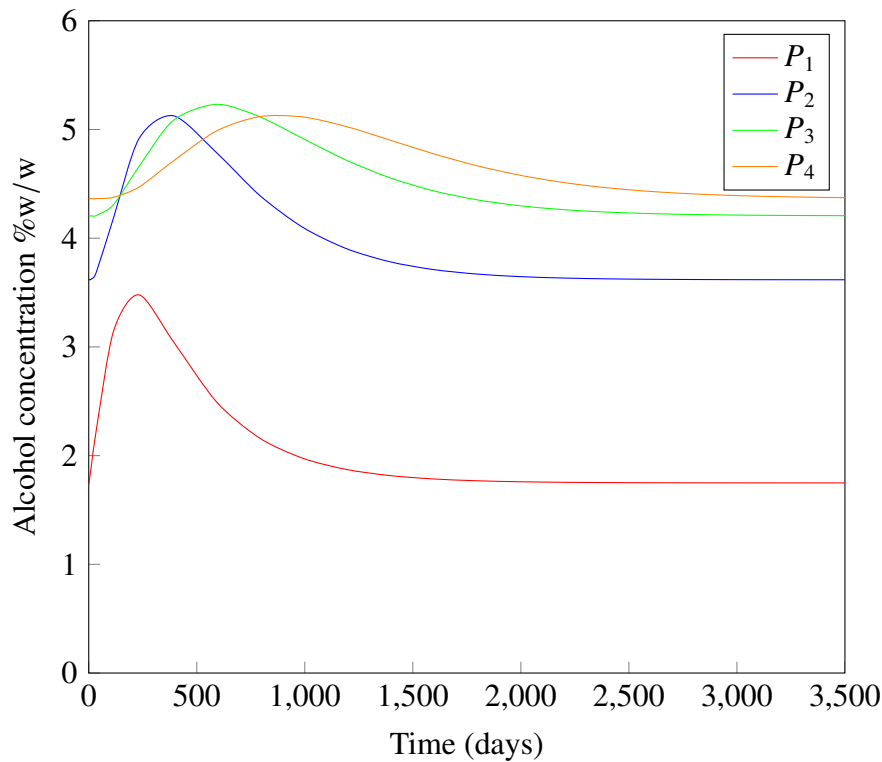


Fig. 6.10. Fermentation response to a 5000 °P pulse signal. Alcohol concentrations.

1.93% when a 10% variation in input materials is introduced. The simulation results can be seen in Figures 6.11, 6.12 and 6.13.

As it can be seen in Figures 6.11, 6.12 and 6.13, the damping effect of the cascade can be proved in the same way as in the mashing stage. However, it can be seen that the damping effect is now much higher than the one observed in mashing. This is mainly due to the huge difference in the dilution rates used in mashing and fermentation. This is more deeply analyzed in Chapter 7, where the design parameters results are discussed.

Finally, the maximum differences of the outputs can be analyzed from the data obtained from the simulation. In this case, the maximum differences of the output variables are, approximately, of 6.22%, 1.86%, and 1.85% for S_4 , X_4 and P_4 respectively.

6.3. Continuous whole brewing system simulation

After analyzing the different subsystems separately, they are simulated linked together hereafter. As is expected that the simulation result of both linked subsystems is not going to provide a very different information to the previous analysis, the purpose is to obtain an overall idea of how the input raw materials, that is, the malt, is affecting the output beer in this whole continuous brewing process.

For doing that, the output signal of the mashing state-space block have been linked to the input of the fermentation model. Of course, the previous fermentation input signal

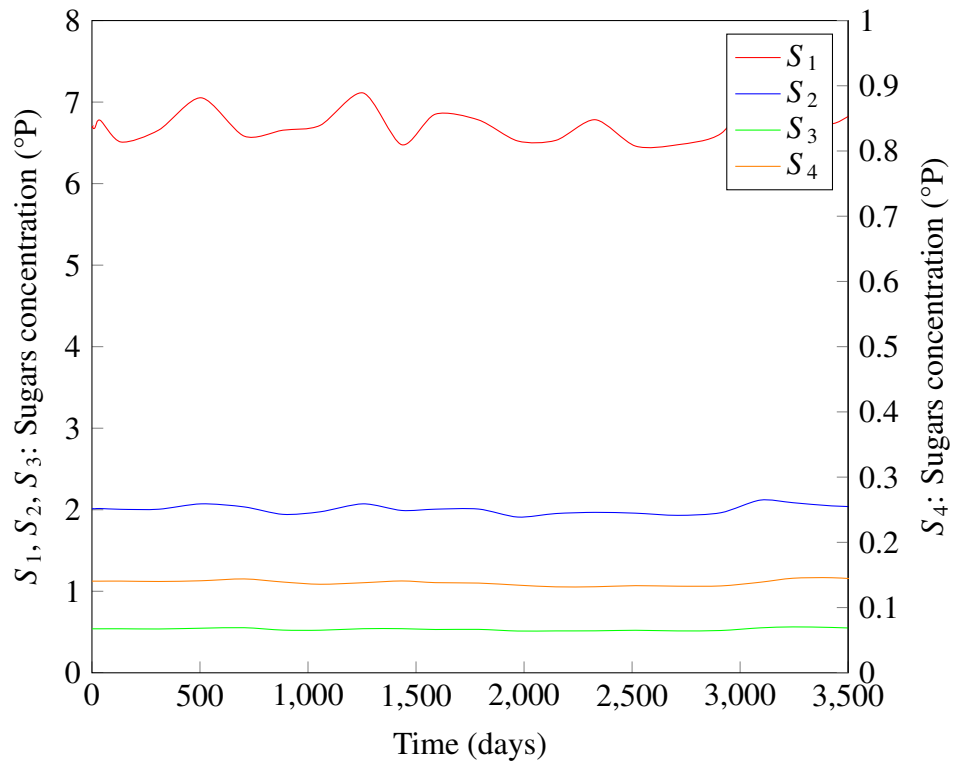


Fig. 6.11. Fermentation response when adding a 10% S_0 amplitude random signal. Sugars concentrations.

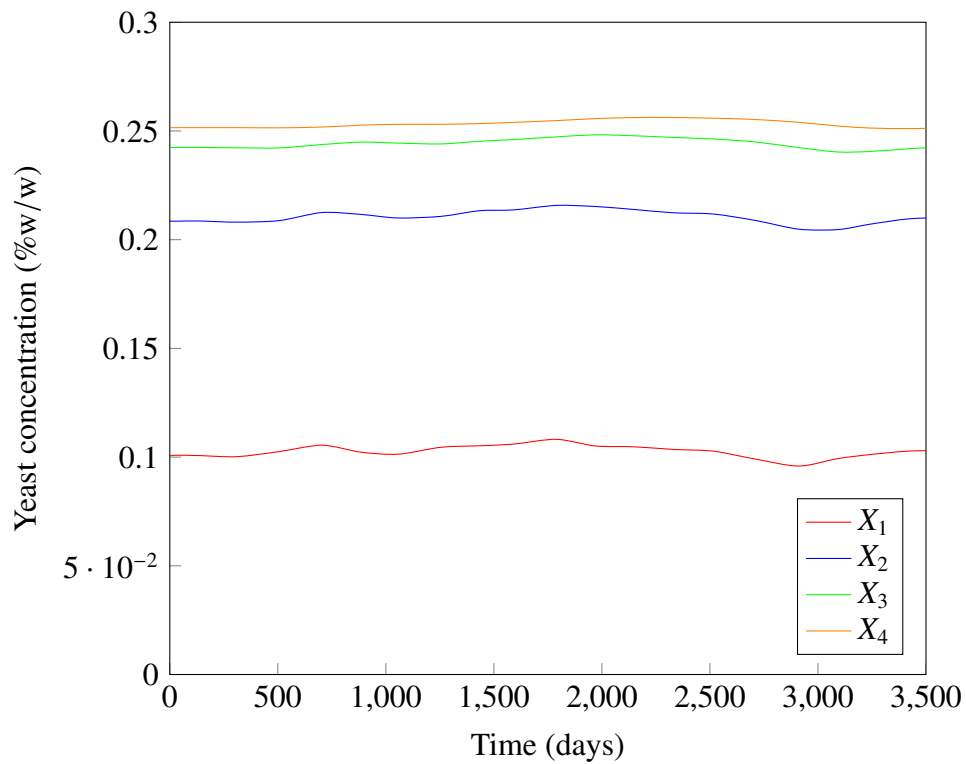


Fig. 6.12. Fermentation response when adding a 10% S_0 amplitude random signal. Yeast concentration.

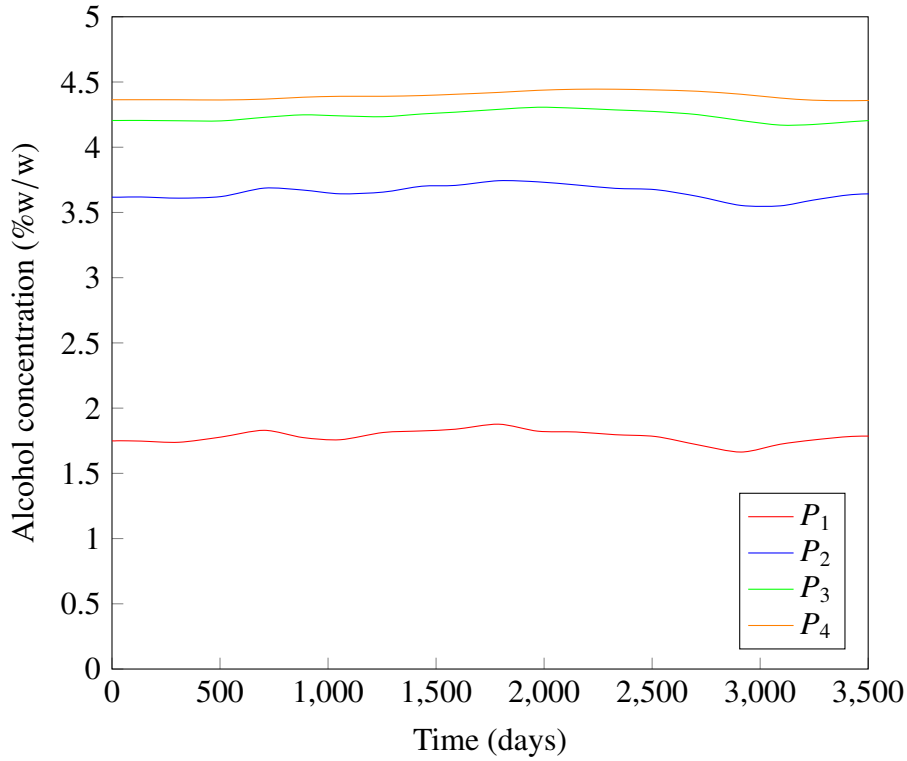


Fig. 6.13. Fermentation response when adding a 10% S_0 amplitude random signal. Alcohol concentration.

has been removed.

It is already known from the isolated subsystems analysis that, by adding a random signal with a 10% difference of the usual input in the mashing system involve to obtain a difference of only 1.93% in the output wort due to the damping effect of the cascade. Also, that the addition of a random signal with 10% difference of the usual sugar concentration that is going into the fermentation cascade, that is, the mashing wort output, is making a difference in the outputs of 6.22%, 1.86%, and 1.85% for S_4 , X_4 and P_4 respectively. Consequently, it is expected that, by linking both subsystems, and adding a random signal to C_{A_0} which varies a 10% of its steady value, with also a 4.32 d^{-1} frequency, there is going to be a negligible effect in the output beer of the whole process. Indeed, the simulation results are exposed in Figure 6.16.

Some interesting information that can be pointed from Figure 6.16 is that, as it was expected, the eight tanks in series (for for mashing and four for fermentation) are making a damping effect against disturbances in amplitude and in time. That is, while the starch concentration can continuously vary around 5% in a very short period of time, the final beer properties do it very little. It is important to highlight that, for showing the effect of the disturbances on the final beer properties, the plots 6.16b, 6.16c, 6.16d and 6.16e are zoomed in amplitude, and much more time is being represented. Hence, the deviations that are shown can be considered negligible. An glimpse of how little the final beer properties can be affected can be proved by pointing that the maximum variations are,

approximately, of 0.07% for S_4 , 0.04% for X_4 , and 0.03% for P_4 .

6.4. Model validation

The last step after modelling and simulation stages may be validation, to test the validity of the model, as "a model is never the true description of a system" [78].

It has to be pointed that, due to the lack of possibility to perform an experimental test, the whole continuous brewing model of this project has not been validated. However, this model is expected to be valid as it has been built with two previously validated and published models that have been adapted from a batch operation mode to a continuous one. Also, as it is showed in Chapter 7, the obtained output wort and beer properties fit the usual values.

Moreover, models are constrained to a certain domain of validity [78]. In the case of this project the domain of validity is expected to be usual brewing operations conditions, as both of the original batch models were built and test with these conditions.

Finally, it has to be mentioned that, due to the limitations of a model, it is necessary to be critical with it. In that way, it is interesting to expose the following three fallacies, briefly compiled from [78]: The Pygmalion effect ("don't fall in love with your model" because the system is more important than it); the Procrustes effect (do not force reality to fit into the model); and to be conscious of the model's accuracy lack.

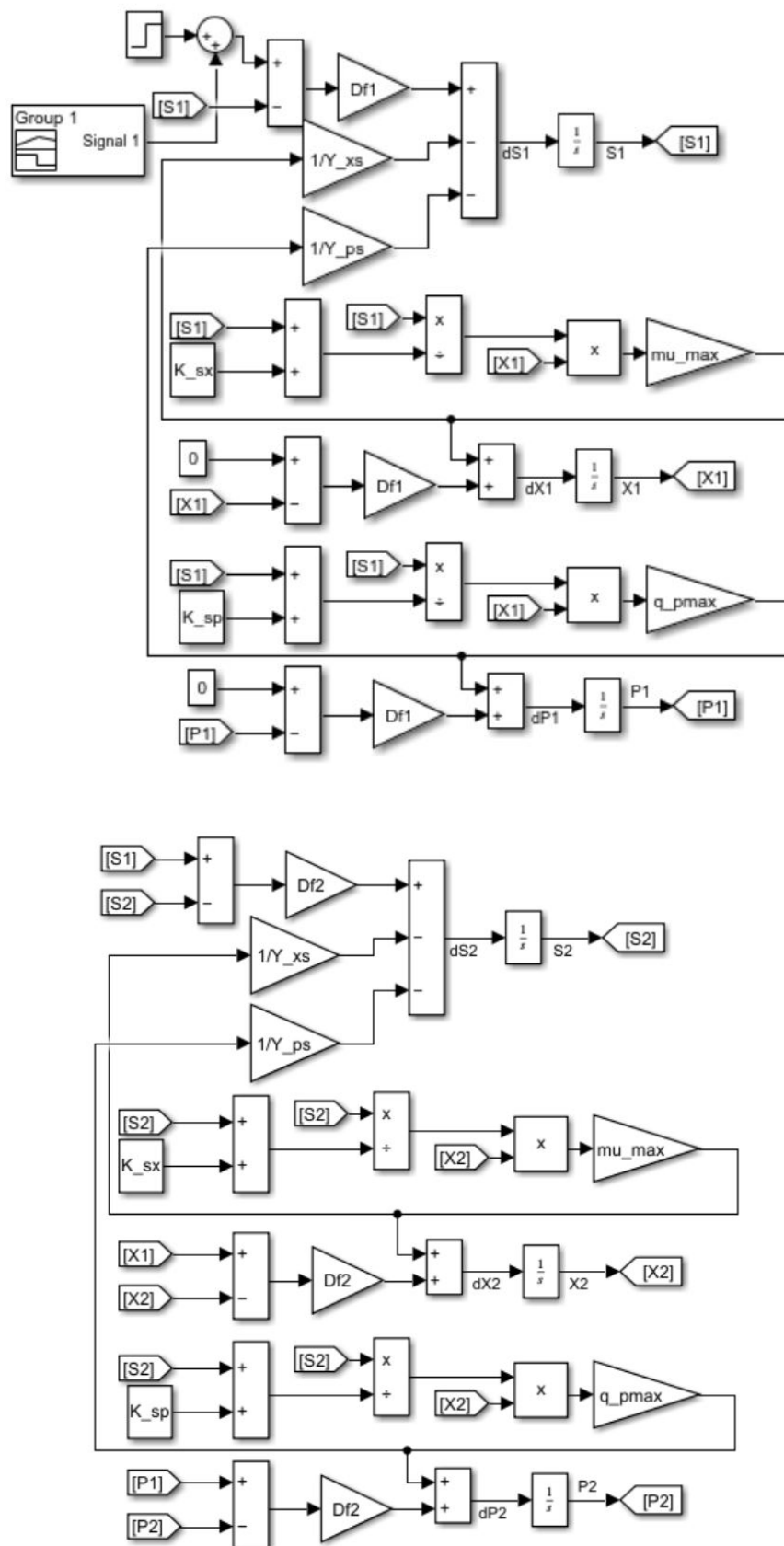


Fig. 6.14. Simulink layout of the fermentation system. Tanks 1 and 2.

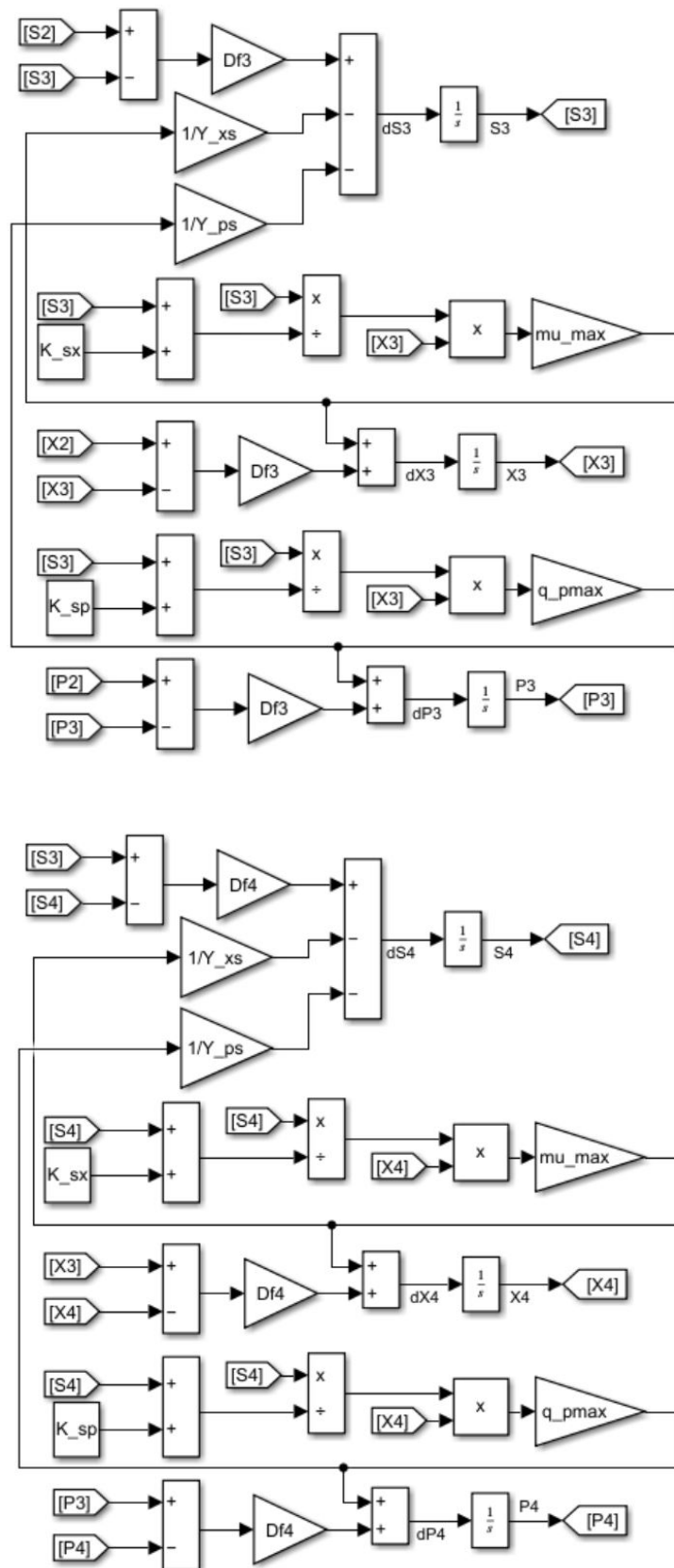
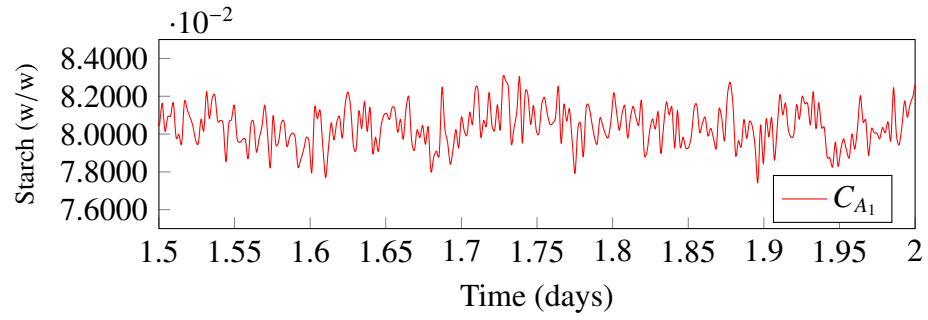
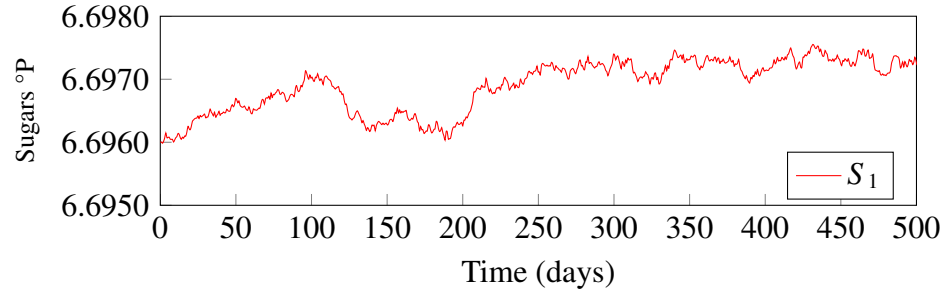


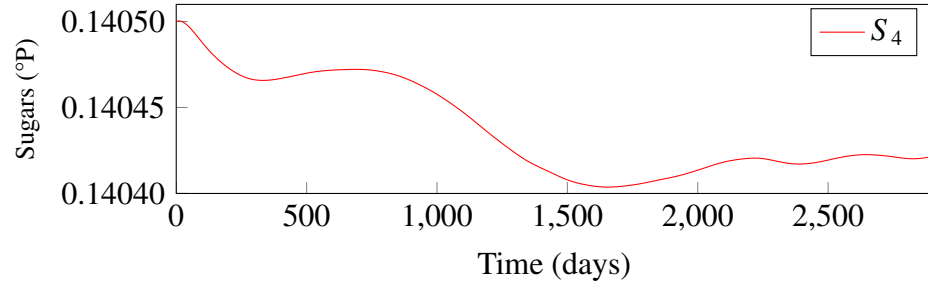
Fig. 6.15. Simulink layout of the fermentation system. Tanks 2 and 3.



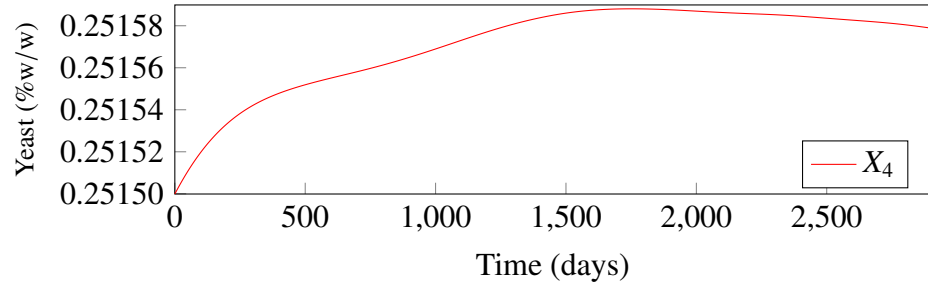
(a) Mashing tank 1 starch concentration.



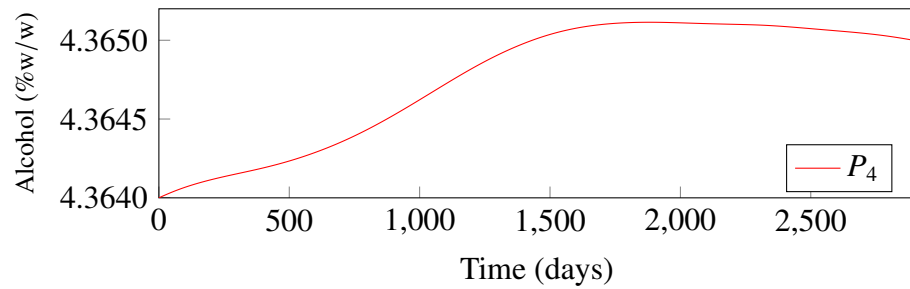
(b) Fermentation tank 1 sugars concentration.



(c) Fermentation tank 4 sugars concentration.



(d) Fermentation tank 1 yeast concentration.



(e) Fermentation tank 1 alcohol concentration.

Fig. 6.16. Whole brewing system response when adding a 10% C_{A0} amplitude random signal.

7. BREWERY DESIGN PARAMETERS RESULTS AND ANALYSIS

One of the main goals of this project is to propose some design parameters for designing a continuous brewing system. Therefore, after an iterative process, Simulation-Based Design have finally provided some suitable parameters for producing a lager beer. All of these parameters are compiled in Table 7.1, and they fit into usual beer properties values, as it is explained hereafter.

KINETIC AND DESIGN PARAMETERS OF THE MODEL	
$E_a = -8814.2\text{J/mol}$	$C_{g2}^{eq} = 9.309^\circ\text{P}$
$R = 8.3144\text{J}/(\text{K mol})$	$C_{g3}^{eq} = 11.071^\circ\text{P}$
$A_{par} = 0.1074^\circ\text{P/K}$ [64]	$C_{g4}^{eq} = 11.080^\circ\text{P}$
$B_{par} = -0.839^\circ\text{P}$ [64]	$D_{f1} = 0.005\text{d}^{-1}$
$C_{par} = -0.132^\circ\text{P/mm}$ [64]	$D_{f2} = 0.003\text{d}^{-1}$
$D_{par} = -90.5^\circ\text{P/w/w}$ [64]	$D_{f3} = 0.003\text{d}^{-1}$
$E_{par} = 45.1^\circ\text{P}$ [64]	$D_{f4} = 0.003\text{d}^{-1}$
$T_{m1} = 318.15\text{K}$	$S_1^{eq} = 6.696^\circ\text{P}$
$T_{m2} = 336.15\text{K}$	$S_2^{eq} = 2.013^\circ\text{P}$
$T_{m3} = 345.15\text{K}$	$S_3^{eq} = 5.397 \times 10^{-1}^\circ\text{P}$
$T_{m4} = 351.15\text{K}$	$S_4^{eq} = 1.405 \times 10^{-1}^\circ\text{P}$
$pH = 5.5$	$X_1^{eq} = 1.008 \times 10^{-1}\%w/w$
$G_r = 4\text{mm}$	$X_2^{eq} = 2.085 \times 10^{-1}\%w/w$
$Di = 0.72w/w$	$X_3^{eq} = 2.424 \times 10^{-1}\%w/w$
$k_0 = 7.52256\text{d}^{-1}$ [64]	$X_4^{eq} = 2.515 \times 10^{-1}\%w/w$
$C_{A0} = 0.1649w/w$	$P_1^{eq} = 1.749\%w/w$
$C_{A_{initial}} = 0.1649w/w$	$P_2^{eq} = 3.617\%w/w$
$D_{m1} = 200\text{d}^{-1}$	$P_3^{eq} = 4.205\%w/w$
$D_{m2} = 70\text{d}^{-1}$	$P_4^{eq} = 4.364\%w/w$
$D_{m3} = 45\text{d}^{-1}$	$\mu_{max} = 0.117\text{d}^{-1}$ [52]
$D_{m4} = 15\text{d}^{-1}$	$q_{pmax} = 2.03\text{g}/(\text{g d})$ [52]
$C_{A1}^{eq} = 8.036 \times 10^{-2}w/w$	$K_{sx} = 150\text{g}/(\text{d m}^3)$ [52]
$C_{A2}^{eq} = 2.283 \times 10^{-2}w/w$	$K_{sp} = 150\text{g}/(\text{d m}^3)$ [52]
$C_{A3}^{eq} = 4.961 \times 10^{-3}w/w$	$Y_{X/S} = 0.093$ [52]
$C_{A4}^{eq} = 4.399 \times 10^{-4}w/w$	$Y_{P/S} = 0.53$ [52]
$C_{g1}^{eq} = 5.139^\circ\text{P}$	

Table 7.1. KINETIC AND DESIGN PARAMETERS OF THE MODEL.

First of all, some parameters values from Table 7.1 might be explained, even if they

were taken from the original model. First, k_0 is appearing originally in [64] as $k_0 = 5.224 \times 10^{-3} \text{min}^{-1}$. It has been converted to fit the fermentation model time units, and make a coherent link. Also, the fermentation kinetic parameters in [52] appear in terms of the beer style and initial wort density. Therefore, the kinetic parameters have been selected for a 11 °P original extract wort, free cells system and bottom fermented yeast, that is, according to a lager style that it is being analyzed, and according to the free cells system that is considered in this project.

Moreover, other considered values have been chosen or adjust to fit into normal brewing values.

The water-mixture dilution rate, Di , can have a value between 1.6 and 3.2 hl per 100 kg of grist for modern mashes according to [63]. However, as the value of Di has to be in w/w, a conversion of the unit have been made to check the range of possible values in weight per weight. Thus, considering water density as 1000 kg/m^3 , the units are converted following Expression 7.1. Moreover, in [64], a value of $Di = 0.75$ is chosen, as it is the average value of the 0.7 to 0.8 range proposed in [79] for Pilsener style production (a lager). Therefore, Di has been properly chosen to fit into these ranges.

$$\frac{1.6 \text{ to } 3.2 \text{ hl water}}{100 \text{ kg grist}} = \frac{160 \text{ to } 320 \text{ kg water}}{260 \text{ to } 420 \text{ kg grist and water}} = 0.61 \text{ to } 0.76 \quad (7.1)$$

The previously justified Di value has an impact to $C_{A_0} = C_{A_{initial}}$ starch concentrations. In [63] is stated that malt contains a 58.9% of starch. Therefore, taking into account that Di is equal to the mass of water divided by the sum of the mass of water and the mass of malt, it is concluded that $C_{A_0} = C_{A_{initial}} = 0.589(1 - Di)$. The final wort extract, $C_{g_4}^{eq}$, fits in the range of wort extract values compiled in [19] of 10.87 to 13.06 °P for a 100% malt Pilsener beer. Regarding the mashing pH, it usually has a value between 5.2 and 5.7 [63]. The value of the grist size has been selected as 4.0 mm, being one of the three studied in the original mashing model, where 4.0 mm, 2.0 mm and 1.19 mm sizes were considered [64].

Finally, regarding the final output beer properties that can be tracked with this model, they also fit with the ranges compiled in [19] for a 100% malt Pilsener beer. The alcohol content fits into the range of 3.05 to 4.66 %w/w (actually, it is almost the suggested mean content of 4.3 %w/w).

The result of S_4^{eq} given in 7.1 is in %w/w, so it has to be converted to %w/v to compare it with the values found in literature. Considering that the specific gravity of lagers are in the range of 1.00682 to 1.01411 [68], the output beer sugar content in equilibrium is, approximately, of $S_4^{eq} \approx 0.15 \text{ %w/v}$. However, the sugar content is lower than the samples exposed in [63], which vary from 0.58 to 0.78 %w/v). Nevertheless, it is not too far from this values, and it could be a good option for diabetic patients. Also, such a low sugar content might not be a problem, as there are other compounds affecting to beer flavour. Anyway, there is still a lot of possible design parameters to change if a higher final sugar

content is desired and, also, it is possible to decrease the final alcohol content and still remain inside the alcohol range for lagers. Finally, this sugar content is likely to achieve, as the continuous fermentation model has been built from a published batch fermentation model.

Although there has not been found any reference to check the usual final yeast concentration, it can be pointed that, after an approximation, the concentrations are similar to the ones that are usually pitched. It is common to pitch 1 gram of dry yeast per liter of wort [17]. Therefore, considering a density of wort of 1000 kg/m^3 as a rough approximation, this would mean that 1 g/L is approximately equivalent to 0.1 %w/w. This value is in the range of the yeast concentrations obtained, taking into account that the ones reached are higher as the yeast population grows.

Regarding the system dynamics results, it can be pointed that the fermentation stage is slower than the mashing, as it is in reality. This obliges to have lower dilution rates in fermentation than in mashing, as fermentation needs more time to perform the reactions. Consequently, the damping effect is much higher in fermentation than in mashing. Moreover, when connecting the two systems in series, the damping effect suffered by the eight tanks becomes really high, making disturbances negligible.

Finally, it has to be reminded some limitations and simplifications of the model. The model is neglecting pipes' length between tanks. Also, only mashing and fermentation steps are considered, because stages between them, as boiling, are not affecting in the compounds reacting in fermentation. These two considerations suggest that the damping effect of a complete brewery may be higher.

8. CONCLUSION AND FUTURE RESEARCH

In this project, a modelling and simulation of a continuous brewing plant have been developed with a Simulation-Based Design approach. Understanding the brewing process first and, specifically, the continuous brewing technologies has been crucial to implement such model. By the adaptation of some published batch mashing and fermentation models into a continuous operation, a linked system has been built. As these models have been previously validated, and the continuous systems are inspired by existing technologies, it is fair to expect that the results of the continuous system are feasible.

8.1. Achievements

After a successful simulation result, that suggest an steady-state continuous operation point, design parameters to build in reality this system have been obtained. Also, the simulation results suggests that the final beer is adjusted to common beer properties.

Moreover, the model gives the possibility of changing some design parameters to adjust the output beer to the desired properties.

8.2. Future research

The first and most important future task to perform would be to carry out an experimental set up following the design parameters proposed in the model, so it could be validated. Moreover, in these experiments, beer flavour properties measures might be carried, as not only the compounds tracked in this project are affecting them.

Then, an adjustment of the model to simulate not only the operational point conditions would be interesting. That is, to suggest a start-up of the brewery plant. To perform an optimized start up, a model in which dilution rates can be varied could be done. This could also test if a variation in beer production could be feasible to face the changing beer demand along a year.

Other actions that can be done is to adjust the model to a possible feedback of beer to previous tanks, to make e thermal model to better understand the temperature influence in the system, to model the yeast settlement or to include other brewing stages, as boiling.

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APPENDIX A: MATLAB SCRIPT

```
1  close all;
2  clc;
3  clear;
4
5  %% MASHING PARAMETERS
6
7  %Parameters:
8  Apar = 0.1074;
9  Bpar = -0.839;
10 Cpar = 0.132;
11 Dpar = -90.5;
12 Epar = 45.1;
13
14 Ea = -8814.2;
15 R = 8.3144;
16
17 Tm1 = 273.15+45;
18 Tm2 = 273.15+63;
19 Tm3 = 273.15+72;
20 Tm4 = 273.15+78;
21
22 pH=5.5;
23 Gr=4;
24 Di=0.72;
25 k0=7.52256;
26 CA0 = 0.589*(1-Di);
27 CAinitial = CA0;
28
29 Dm1 = 200;
30 Dm2 = 70;
31 Dm3 = 45;
32 Dm4 = 15;
33
34 k1 = k0*exp(-Ea/(R*Tm1));
35 k2 = k0*exp(-Ea/(R*Tm2));
36 k3 = k0*exp(-Ea/(R*Tm3));
37 k4 = k0*exp(-Ea/(R*Tm4));
38
39 %Simplifications:
40 W1 = (k1/CAinitial)*(Apar*Tm1+Bpar*pH+Cpar*Gr+Dpar*Di+Epar);
41 W2 = (k2/CAinitial)*(Apar*Tm2+Bpar*pH+Cpar*Gr+Dpar*Di+Epar);
42 W3 = (k3/CAinitial)*(Apar*Tm3+Bpar*pH+Cpar*Gr+Dpar*Di+Epar);
43 W4 = (k4/CAinitial)*(Apar*Tm4+Bpar*pH+Cpar*Gr+Dpar*Di+Epar);
44
45 %State-space matrices:
```

```

46  Assm = [-Dm1-k1 0 0 0 0 0 0 0;
47          Dm2 -Dm2-k2 0 0 0 0 0 0;
48          0 Dm3 -Dm3-k3 0 0 0 0 0;
49          0 0 Dm4 -Dm4-k4 0 0 0 0;
50          W1 0 0 0 -Dm1 0 0 0;
51          0 W2 0 0 Dm2 -Dm2 0 0;
52          0 0 W3 0 0 Dm3 -Dm3 0;
53          0 0 0 W4 0 0 Dm4 -Dm4];
54  Bssm = [Dm1; 0; 0; 0; 0; 0; 0; 0];
55  Cssm = [0 0 0 0 0 0 0 1];
56  %Cssm = eye(8); %Uncomment to track all the state variables
57  Dssm = 0;
58  %Dssm = [0;0;0;0;0;0;0;0]; %Uncomment to track all the state
    variables
59
60  e_m = eig(Assm);
61
62  %Initial conditions:
63  xssm0 =
    [0.08036;0.02283;0.004961;0.0004399;5.139;9.309;10.71;11.08];
64
65
66  %% FERMENTATION
67
68  % Parameters:
69  Df1=0.005;
70  Df2=0.003;
71  Df3=0.003;
72  Df4=0.003;
73
74  S1_eq=6.696;
75  X1_eq=0.1008;
76  P1_eq=1.749;
77
78  S2_eq=2.013;
79  X2_eq=0.2085;
80  P2_eq=3.617;
81
82  S3_eq=0.5397;
83  X3_eq=0.2424;
84  P3_eq=4.205;
85
86  S4_eq=0.1405;
87  X4_eq=0.2515;
88  P4_eq=4.364;
89
90  mu_max=0.117;
91  K_sx=150;
92  K_sp=150;
93  q_pmax=2.03;
94  Y_xs=0.093;

```

```

95 Y_ps=0.53;
96
97 % Simplifications:
98 Q1=mu_max*((K_sx+S1_eq)*X1_eq-S1_eq*X1_eq)/(K_sx+S1_eq)^2;
99 T1=mu_max*S1_eq/(K_sx+S1_eq);
100 U1=q_pmax*((K_sp+S1_eq)*X1_eq-S1_eq*X1_eq)/(K_sp+S1_eq)^2;
101 V1=q_pmax*S1_eq/(K_sp+S1_eq);
102
103 Q2=mu_max*((K_sx+S2_eq)*X2_eq-S2_eq*X2_eq)/(K_sx+S2_eq)^2;
104 T2=mu_max*S2_eq/(K_sx+S2_eq);
105 U2=q_pmax*((K_sp+S2_eq)*X2_eq-S2_eq*X2_eq)/(K_sp+S2_eq)^2;
106 V2=q_pmax*S2_eq/(K_sp+S2_eq);
107
108 Q3=mu_max*((K_sx+S3_eq)*X3_eq-S3_eq*X3_eq)/(K_sx+S3_eq)^2;
109 T3=mu_max*S3_eq/(K_sx+S3_eq);
110 U3=q_pmax*((K_sp+S3_eq)*X3_eq-S3_eq*X3_eq)/(K_sp+S3_eq)^2;
111 V3=q_pmax*S3_eq/(K_sp+S3_eq);
112
113 Q4=mu_max*((K_sx+S4_eq)*X4_eq-S4_eq*X4_eq)/(K_sx+S4_eq)^2;
114 T4=mu_max*S4_eq/(K_sx+S4_eq);
115 U4=q_pmax*((K_sp+S4_eq)*X4_eq-S4_eq*X4_eq)/(K_sp+S4_eq)^2;
116 V4=q_pmax*S4_eq/(K_sp+S4_eq);
117
118 % State-space matrices:
119 Assf = [(-Df1-(Q1+U1)/Y_xs) 0 0 0 (-(T1+V1)/Y_ps) 0 0 0 0 0 0 0;
120         Df2 (-Df2-(Q2+U2)/Y_xs) 0 0 0 (-(T2+V2)/Y_ps) 0 0 0 0 0 0;
121         0 Df3 (-Df3-(Q3+U3)/Y_xs) 0 0 0 (-(T3+V3)/Y_ps) 0 0 0 0 0;
122         0 0 Df4 (-Df4-(Q4+U4)/Y_xs) 0 0 0 (-(T4+V4)/Y_ps) 0 0 0 0;
123         Q1 0 0 0 (-T1-Df1) 0 0 0 0 0 0 0;
124         0 Q2 0 0 0 (-T2-Df2) 0 0 0 0 0 0;
125         0 0 Q3 0 0 0 (-T3-Df3) 0 0 0 0 0;
126         0 0 0 Q4 0 0 0 (-T4-Df4) 0 0 0 0;
127         U1 0 0 0 V1 0 0 0 -Df1 0 0 0;
128         0 U2 0 0 0 V2 0 0 Df2 -Df2 0 0;
129         0 0 U3 0 0 0 V3 0 0 Df3 -Df3 0;
130         0 0 0 U4 0 0 0 V4 0 0 Df4 -Df4];
131 Bssf = [Df1; 0; 0; 0; 0; 0; 0; 0; 0; 0; 0; 0];
132 Cssf = [0 0 0 1 0 0 0 0 0 0 0 0;
133         0 0 0 0 0 0 0 1 0 0 0 0;
134         0 0 0 0 0 0 0 0 0 0 0 1];
135 %Cssf = eye(12); %Uncomment to track all the state variables
136 Dssf = 0;
137 %Dssf = [0; 0 ;0; 0; 0; 0; 0; 0; 0; 0; 0; 0]; %Uncomment to track
    all the state variables
138
139 %Initial conditions:
140 xssf0 = [S1_eq; S2_eq; S3_eq; S4_eq; X1_eq; X2_eq; X3_eq; X4_eq;
    P1_eq; P2_eq; P3_eq; P4_eq];

```